

Chapter One

INTRODUCTION to Operations Management

Mulugeta G.

Sheba University College

Faculty of Business and Economics

Department of Management

Definition of Operations Management

- ▶ Operations is responsible for supplying the *product* or *service* of the organization.
- ▶ Managers are involved in planning, organizing and controlling; and Operation Manager has direct responsibility of “getting the job done”
- ▶ Operation Manager’s job is to manage the process of converting inputs into desired outputs.
- ▶ Operations managers make *decisions* regarding the operations function and its *connection* with other *functions*.

Cont...

- ▶ Operations managers *plan, organize & control* the production *process* and its interfaces within the organization and with the external environment.
- ▶ OM is about getting the day-to-day work done quickly, efficiently, without error, & at low cost. OM creates dramatic improvements in customers services and reduction in cost
- ▶ OM defined as the design, operation and improvement of the systems that create and deliver the firm's primary products and services
- ▶ OM is concerned with the management of entire system (from input to output) that produces goods or deliver products. (deals with effective and efficient transformation process)

Key Points in OM Definition

- Operation Management is the field of study that tries to understand, explain, predict and change organizational and strategic effects of the transformation process.
- Hence, OM deals with the effective and efficient management of the transformation process
- *Decisions* : the operations manager must make *Decisions (process, quality, capacity, and inventory)*
- *Functions* : *Operation is the main functions* in the organization. (manufacturing, Production dept)
- *Process*: *Plan and control Process* for producing goods and services

Major Decisions Framework for OM

- Process
 - ✓ How to produce & deliver
- Quality
 - ✓ Criteria, measurement & process for achieving
- Capacity
 - ✓ Physical facilities & labor
- Inventory
 - ✓ What, when & how much?

Examples of Important Policies in Operations

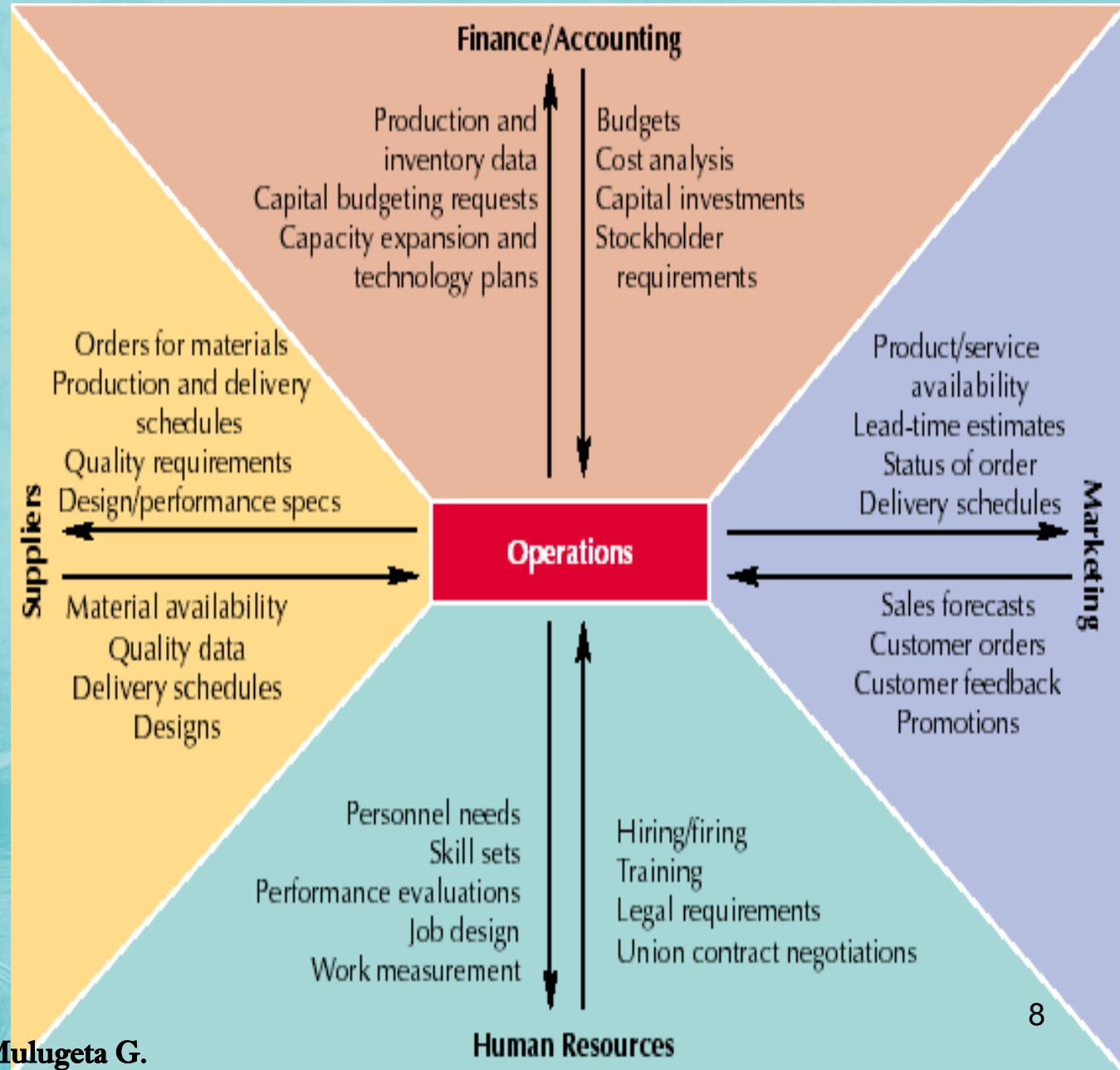
<i>Policy Type</i>	<i>Policy Area</i>	<i>Strategic Choices</i>
Process	Span of process	Make or buy
	Automation	Handmade or machine-made
	Process flow	Flexible or specialized
	Job specialization	Project, batch, line, or continuous
	Supervision	Centralized or empowered workers
Quality Systems	Approach	Prevention or inspection
	Training	Technical or managerial training
	Suppliers	Selected on quality or cost
Capacity	Facility size	One large or several small facilities
	Location	Near markets, labor, or materials
	Investment	Permanent or temporary
Inventory	Amount	High or low levels of inventory
	Distribution	Centralized or decentralized warehouses
	Control Systems	Control in great detail or less detail

Cross-Functional Decision Making

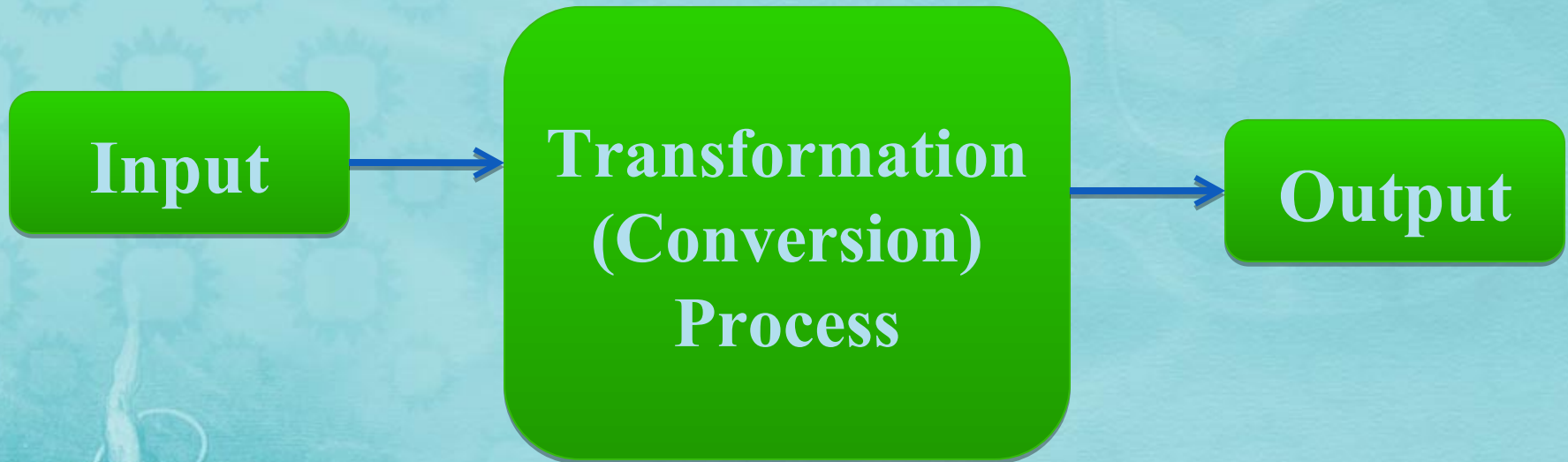
- Operations as the primary function: OM is a functional field of business with clear line management responsibilities
 - ✓ Uses decision making tools
 - ✓ Considers facility automation
- Other primary functions:
 - ✓ Marketing
 - ✓ Finance
 - ✓ Human (manpower) ...
- Other supporting functions may exist

Operations Function

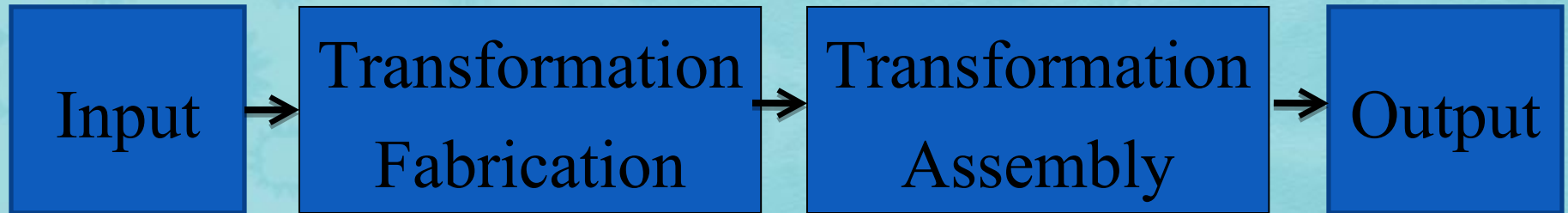
- Operations
- Marketing
- Finance and Accounting
- Human Resources
- Outside Suppliers



Operations as a System (Process)



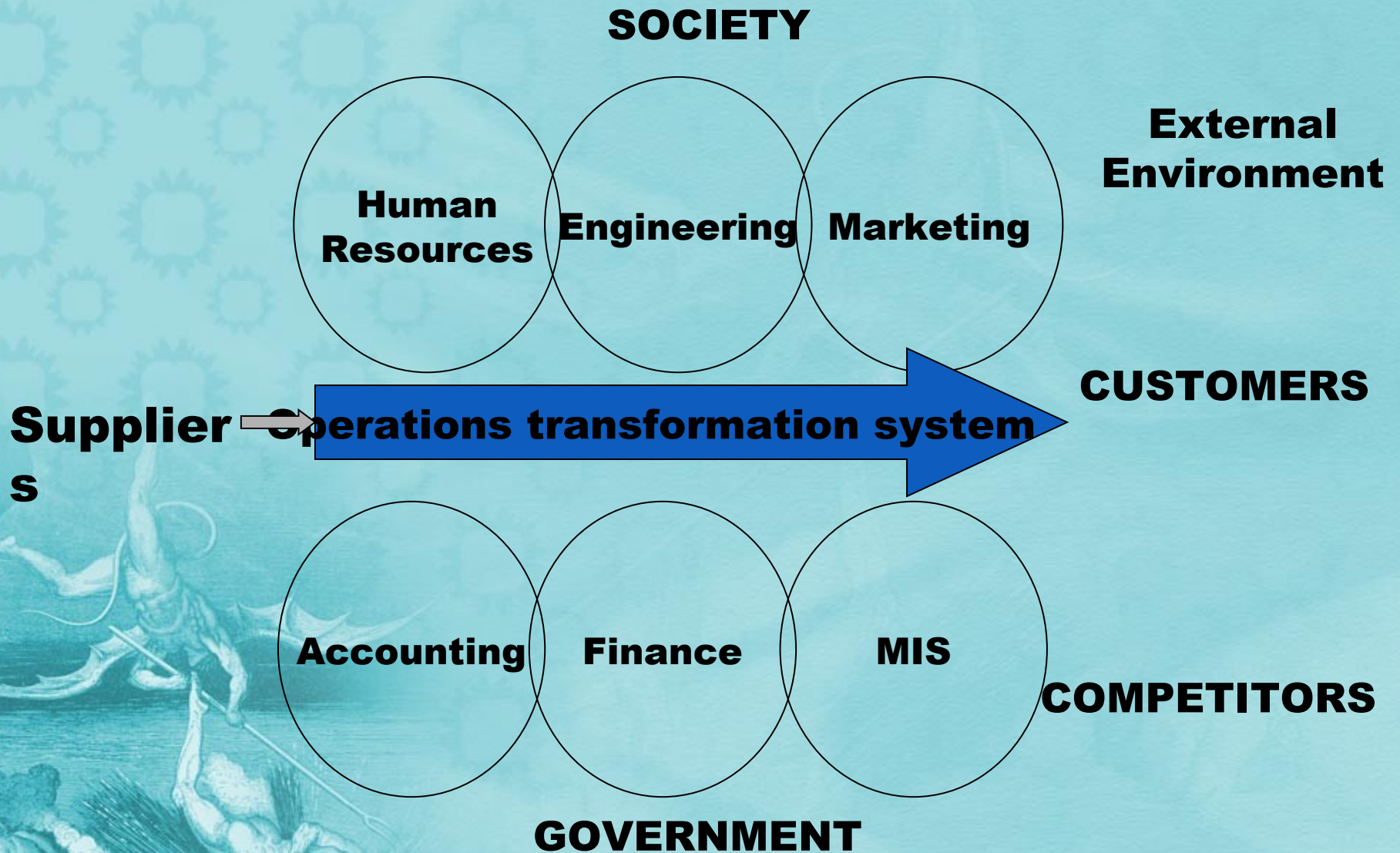
Operations as a Process



Fabrication: making the parts

Assembly: putting the parts together

Relation of Operations to its Environment



Difference b/n Manufacturing & Service Operation

- Tangible/intangible nature of output
- Can be inventoried/ can not be inventoried
- Consumption of outputs (consume overtime, consumes immediately Simultaneous consumption and production)
- Nature of the work (less labor and more equipment, high labor and less equipment)
- Degree of customer contact
 - ✓ little consumer contact, Extensive consumer contact
- Customer participation in conversion
 - ✓ distinguish b/n output (generated service, medical service) & throughput (going through the processes)
- Measurement of performance/quality easy to assess/
difficulty to assess

Key Element of Operation Management

- × Product selection and design
- × Process selection and planning
- × Facilities (plant) location, Facilities layout and material handling
- × Capacity planning
- × Production planning and control (PPC)
- × Inventory control
- × Quality assurance and control
- × Work study and job design
- × Cost reduction and cost control

Contemporary & Core Issues Operations Themes

- Service and Manufacturing (differences and implications)
- Customer-Directed Operations
- Time Reduction (Lean Operations)
- Integration of Operations and Other Functions
- Environmental Concerns
- Supply Chain Management
- Globalization of Operations

Productivity

- Productivity: Output/Input
- Total measure Productivity = $\frac{\text{Outputs}}{\text{Inputs}}$
or
= $\frac{\text{Goods and services produced}}{\text{All resources used}}$
- Partial measures of productivity =
 $\frac{\text{Output}}{\text{Labor}}$ or $\frac{\text{Output}}{\text{Capital}}$ or $\frac{\text{Output}}{\text{Materials}}$ or $\frac{\text{Output}}{\text{Energy}}$

Cont...

- Multifactor measures of productivity =

- $$\frac{\text{Output}}{\text{Labor} + \text{Capital} + \text{Energy}}$$

or

- $$\frac{\text{Output}}{\text{Labor} + \text{Capital} + \text{Materials}}$$

Example of Productivity Measurement

- MIE is compiling the monthly productivity report for its BOD. The average labor rate is Birr 15 per hour, and the average machine usage rate is Birr 10 per hour.

Description	Amount
Units produced	100,000
Labor hours	10,000
Machine hours	5,000
Cost of materials	Birr 35,000
Cost of energy	Birr 15,000

Solution

(a) Labor productivity = output/labor hour
$$= 100,000/10,000 = 10 \text{ units /hours}$$

(b) Machine productivity = output/machine hours
$$= 100,000/5,000 = 20 \text{ units /hour}$$

(c) Multifactor Productivity = Output/ labor costs+ Machine costs+ Material costs +Energy costs
$$= 100,000 / (10,000 \times \text{Br}.15) + (5,000 \times \text{Br}10) + \text{Br}35,000 + \text{Br}.15,000$$

$$= 100,000 / \text{Br}250,000 = 0.4 \text{ unit per a Birr spent}$$

Chapter Two

Operations Strategy

Mulugeta G.

**Sheba University College,
Faculty of Business and Economics,
Department of Management**

What is Strategy ?

- **Strategy** is how the mission of a company is accomplished .
- It unites an organization.
- It provides consistency in decisions, and keeps the organization moving in the right direction

Processes of Strategy Formulation

1. Environmental scanning
2. Internal Environment Analysis
3. Development of corporate level strategies
4. Development of business level strategies
5. Development of Functional level strategies

This includes studying the nature of :

- ✓ the industry, the market place , the competitors , the legal and political environment, technology, the nature of competitors
- ✓ The sources and power of influence
- ✓ Identifying the business's possible opportunities and threats from the environment (internal and external)

Internal Environment Analysis

This involves:

- ✓ Identifying the firm's resources and defining their weaknesses and strengths
- ✓ Identifying the firm's possible sources of capabilities
- ✓ Defining the firm's possible comparative advantages
- ✓ Defining the competitive advantages of the business

Corporate and Business Level Strategies

Corporate Level Strategies

- Expansion
- Stability
- Retrenchment strategies

Business Level Strategies

- Cost leadership
- Differentiation strategies
- Market niche strategies

Functional Level Strategies

- Marketing strategies
- Production Strategies
- Finance Strategies
- Logistics Strategies etc.

End Results of the Strategic Formulation

- At the end of the strategy formulation, the firm should be able to:
 1. **Defining its primary task**-this is the purpose of the firm- **what the firm is in the business of doing** . e.g. Ethiopian Airlines is in the business of transportation
 2. **Assessing core competencies** (distinctive competence)- is what a firm does better than any one else
 3. **Determining order winners and order qualifiers.**
 - ✓ This helps to determine what customers influence their purchase decision.
 - **Order qualifiers**- are the characteristics of a product or service that qualify it to be considered for purchase by a customer. E.g. brand name , reliability, design features, gas mileage
 - **Order winner**- is the characteristics of a product or service that wins orders in the marketplace –**the final factor in the purchasing decision**. E.g. price, quality, after sales service.

Cont...

Example:

- A brand name car can be an “order qualifier”
- Others like reliability, design features, gas mileage

Repair services can be “order winners”

Examples:

Warranty, roadside Assistance, Leases, etc.



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Cont...

- Positioning the Firm- No firm can be all things to all people !!!
- Hence a firm should choose one or two important things on which to concentrate and doing them extremely well- this is named as **strategic positioning**.
- A firm's positioning strategy defines how it will compete in the market place –what unique value it will deliver to the customer.

E.g. competing on cost, quality, flexibility, speed

Operations Strategy

- **Operations strategy** is defined as the set of decisions that are warranted in the operational processes in order to support the competitive strategies of the business.
- Operations strategy is designed to give the firm a **competitive advantages** in the products or services that are served to the customers .
- It is the **road map** of the operations function
- Operations strategy decisions include:
 - ✓ The **capacity** decisions
 - ✓ The type of processes and manufacturing **technologies** to be adopted in the production processes
 - ✓ The **nature** of products to be produced etc

Operations Strategy Development Processes

1. Developing operations **mission**
2. Developing operations **objectives**
3. Developing operations **priorities (core competencies)**
4. Developing operations **policies**

Operations Objectives

- Usually the operations Objectives are :
 - ✓ Cost
 - ✓ Quality
 - ✓ Delivery
 - ✓ Flexibility

Cont...

- **Cost**- this is the ability to reduce the costs of production.
 - ✓ This can include reducing cost of labor, material, overhead expenses etc.
 - ✓ This can ensure lower price and higher profitability
- **Quality**-this includes higher product performance, superior features, greater durability, convenient services
- **Time /Delivery**- faster delivery time, on-time delivery, speedy product development , reliable delivery.
- **Flexibility**- this is the ability to provide a wide variety of products , quick changes in volume of production
- This can include:
 - ✓ Customization
 - ✓ Volume flexibility

Core Competencies

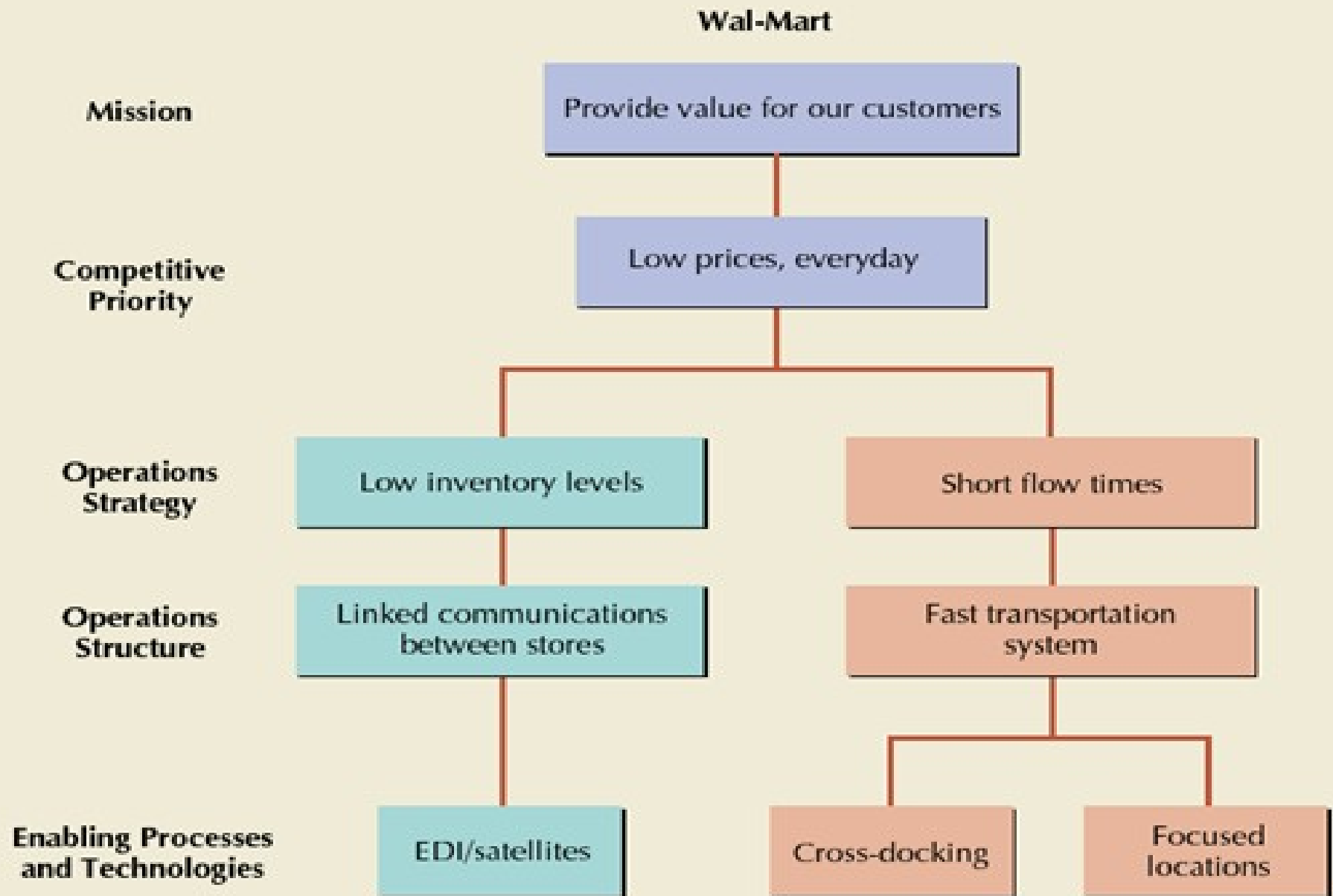
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- Core competencies are the **unique capabilities of the business organization apart from its competitors** .
- These competitive competencies should not be copied and imitated by others . E.g. the **lowest price, the best quality, highly flexible production system**

Elements /components of Operations Strategy

1. Designing of the production system
 - ✓ Customized product design
 - ✓ Standard product design
2. Facilities for production and services
3. product or service design and development
4. technology selection, development and process dev't
5. Allocation of resources
6. Focus on facilities planning

Operations Strategy at Wal-Mart



Strategic Decisions in Operations

❑ Products and Services

- Make-to-stock
- Assemble-to-order
- Make-to-order

❑ Processes & Technology/Production Systems

- Project type
- Batch production
- Mass production
- Continuous Production
- Assembly Production

Cont...

☐ Human Resources

- Skill levels and degree of autonomy required in production
- Selection criteria and training requirements
- Setting up policies on performance evaluations
- Compensation and incentives

☐ Quality

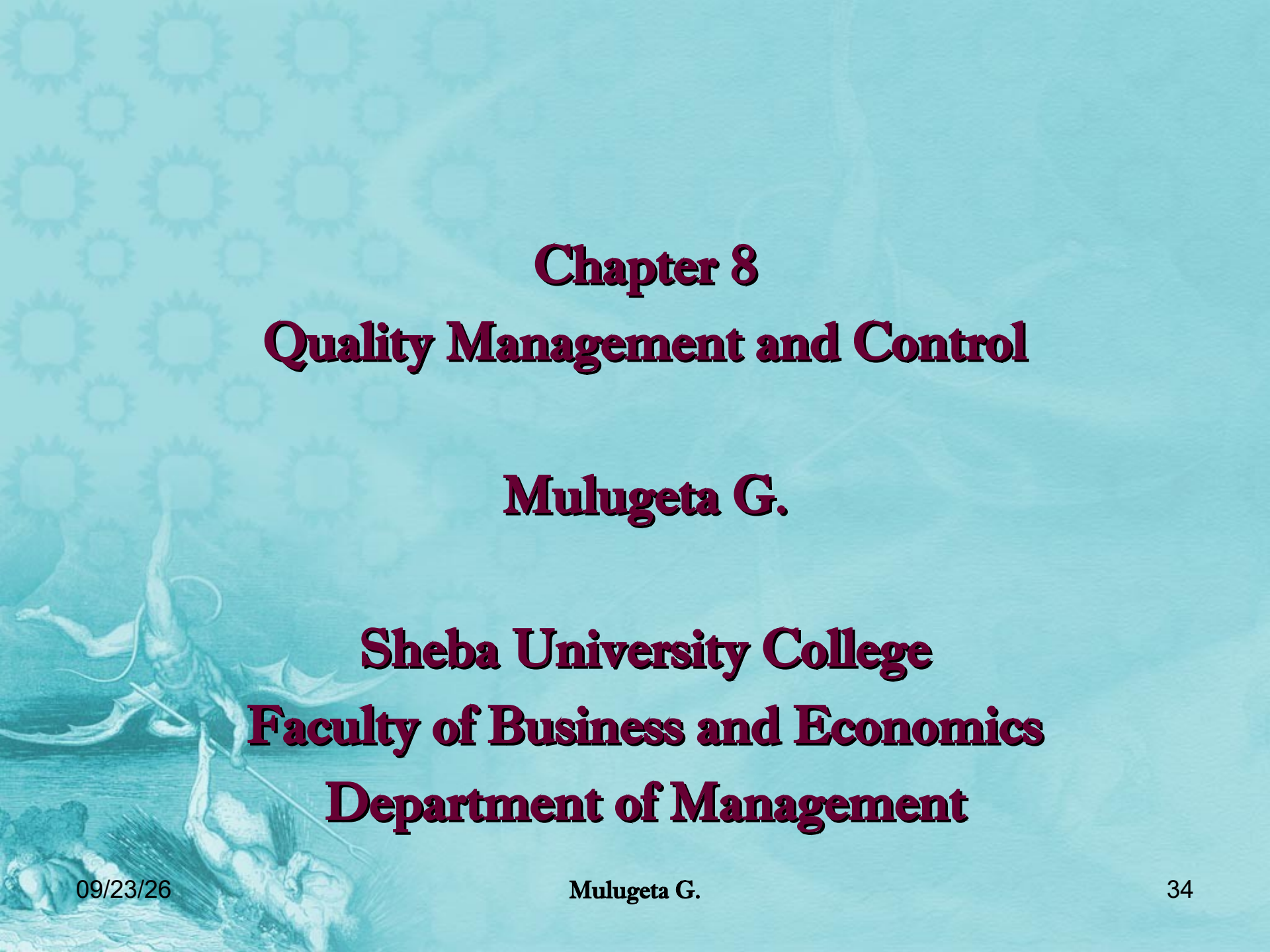
- What is the target level of quality for the products /services ?
- How will it be measured?
- How will employees be involved with quality?

Cont...

❑ **Sourcing** –decision on make or buy decision

- On what basis should particular items be made in house?
- When should items be outsourced?
- How should suppliers be selected ?
- How many suppliers should be used?





Chapter 8

Quality Management and Control

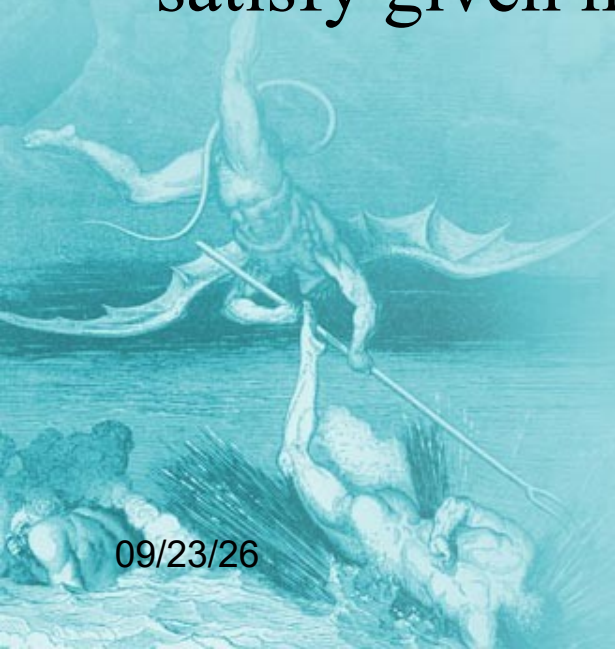
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What is Quality?

Quality is like beauty !!!

- ❑ Meeting, or exceeding, customer requirements now and in the future.
- ❑ Quality is the totality of features and characteristics of a product or service that bears on its ability to satisfy given needs.



Meaning of Quality: Consumer's Perspective

- **Fitness for use**

- ✓ how well product or service does what it is supposed to

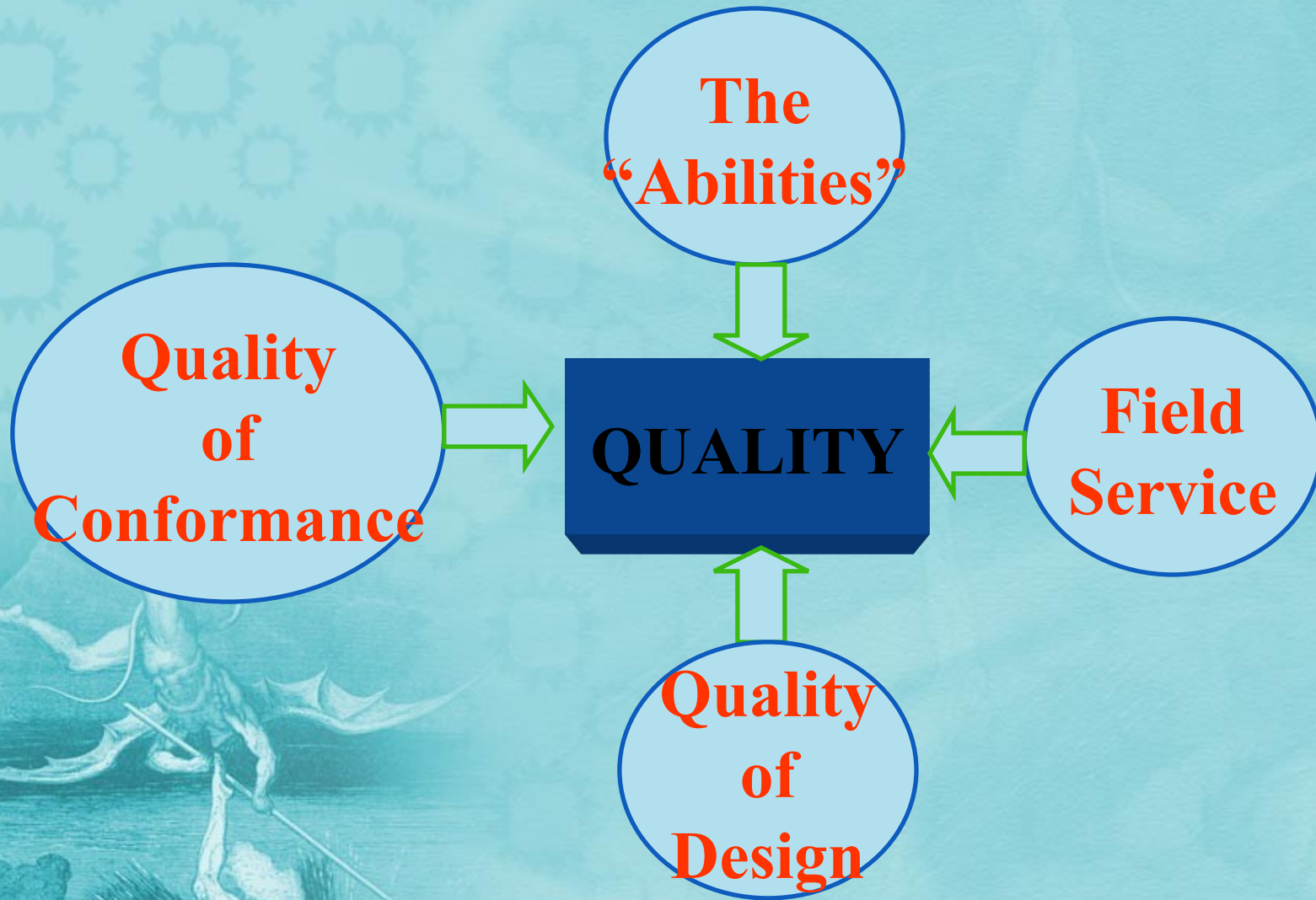


- **Quality of design**

- ✓ designing quality characteristics into a product or service
- ✓ A Mercedes and a Ford are equally “fit for use,” but with different design dimensions



Dimensions of Quality in Manufacturing



Quality of Design

- Determined before the product is produced
- Determined by market research
- Translates the “wishes” of customers into specifications

Quality of Conformance

Producing a product that meets the specifications

Abilities

- **Availability** (Continuity of service to customers)
- **Reliability** (Length of time that a product can be used before it fails)
- **Maintainability** (Restoration of the product or service once it has failed)

Dimensions of Quality in Services

1. **Time and Timelines:** How long must a customer wait for service and is it completed on time ?
2. **Completeness:** Is everything the customer asked for provided?
3. **Courtesy:** How are customers treated by employees?
4. **Consistency:** is same level of service provided to each customer each time?
 - **Accessibility and convenience**
 - ✓ how easy is it to obtain service?
 - ✓ does service representative answer you calls quickly?
 - **Accuracy**
 - ✓ is service performed right every time?
 - ✓ is your bank or credit card statement correct every month?
 - **Responsiveness**
 - ✓ how well does company react to unusual situations?
 - ✓ how well is a telephone operator at able to respond to a customer's questions?

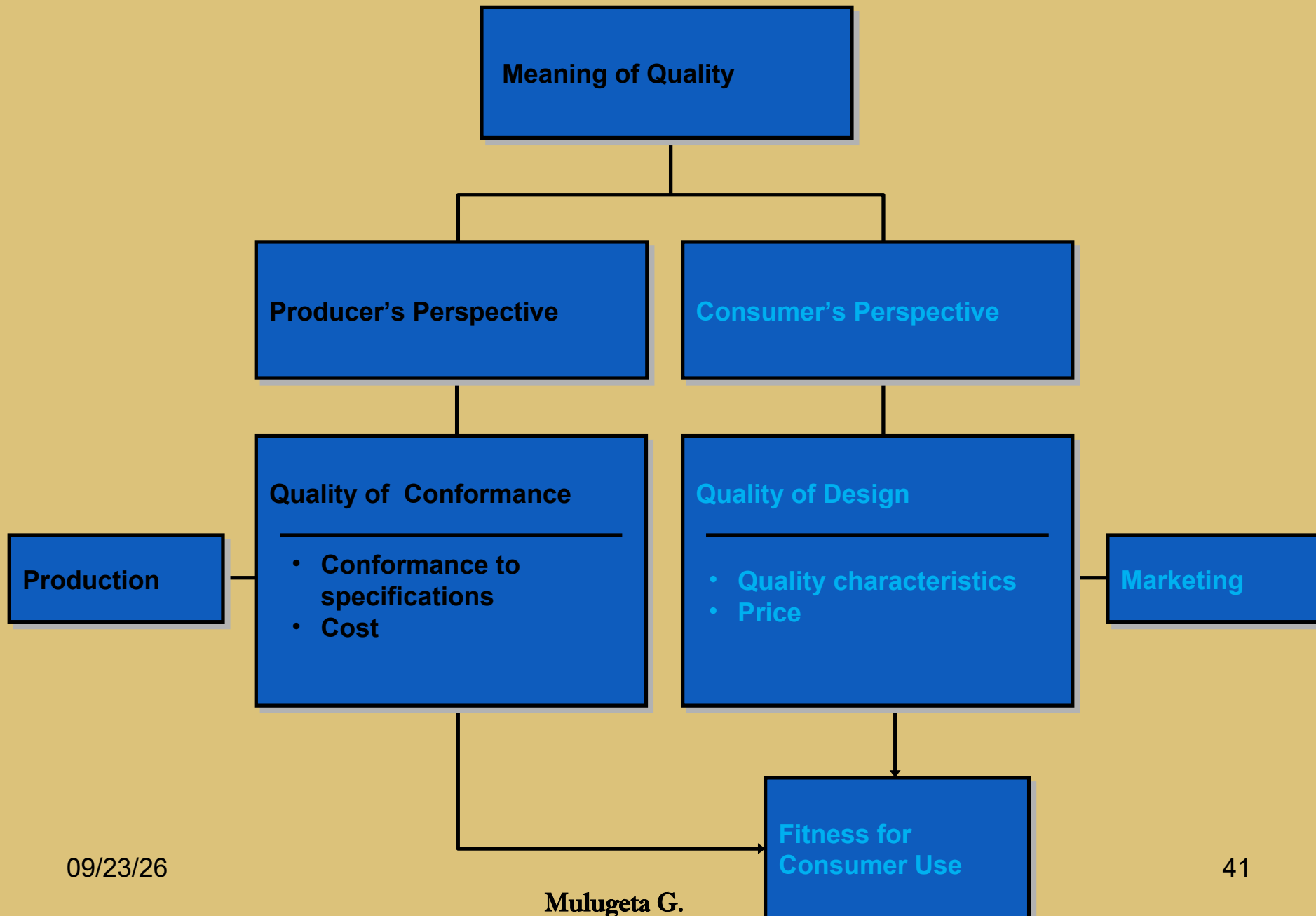
Meaning of Quality: Producer's Perspective

Quality of conformance

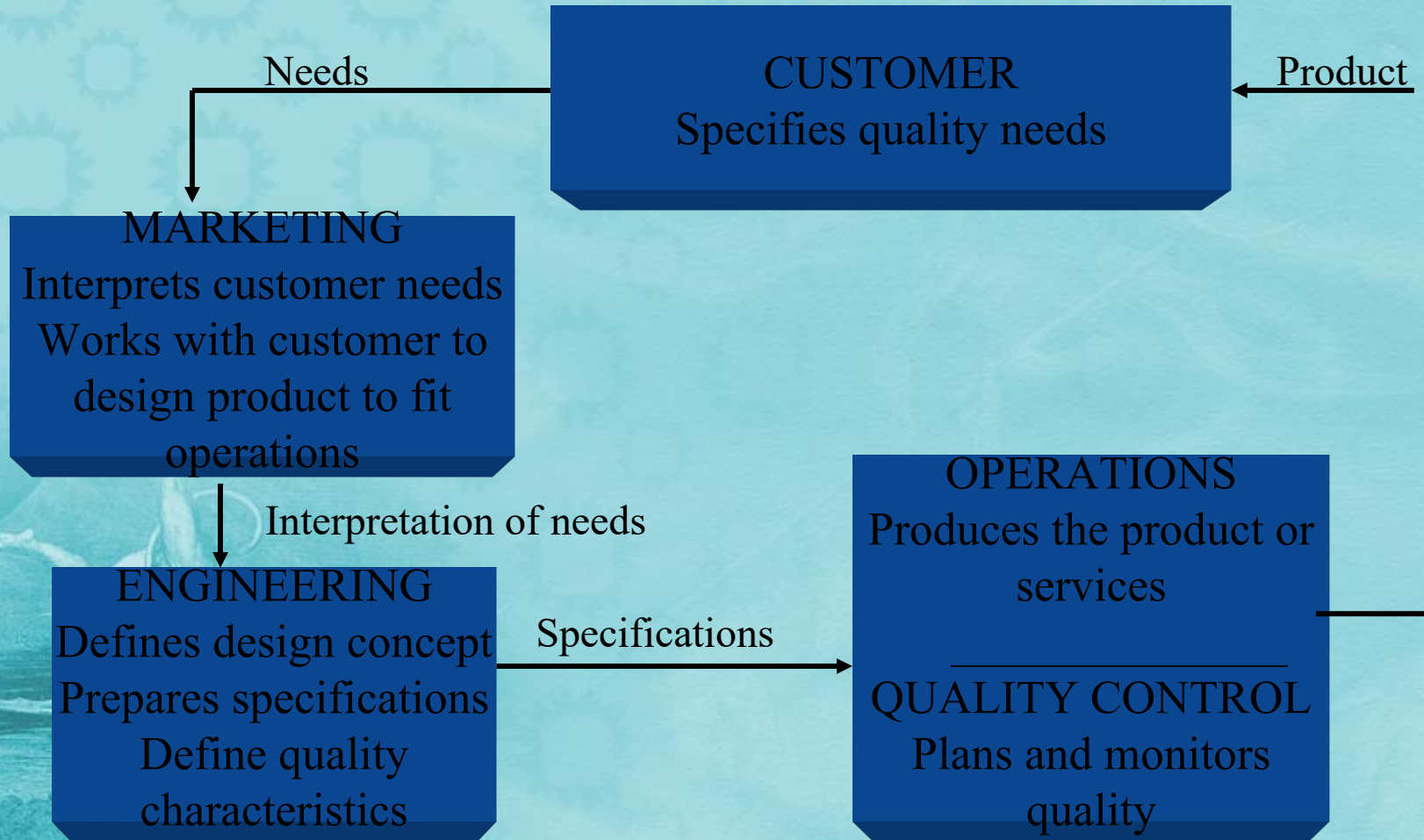
- Making sure product or service is produced according to design
 - if new tires do not conform to specifications, y wobble
 - if a hotel room is not clean when a guest checks in, hotel is not functioning according to specifications of its design

Meaning of Quality: A Final (customer) Perspective

- Consumer's and producer's perspectives depend on each other
- Consumer's perspective: PRICE
- Producer's perspective: COST
- Consumer's view must dominate



Quality Cycle

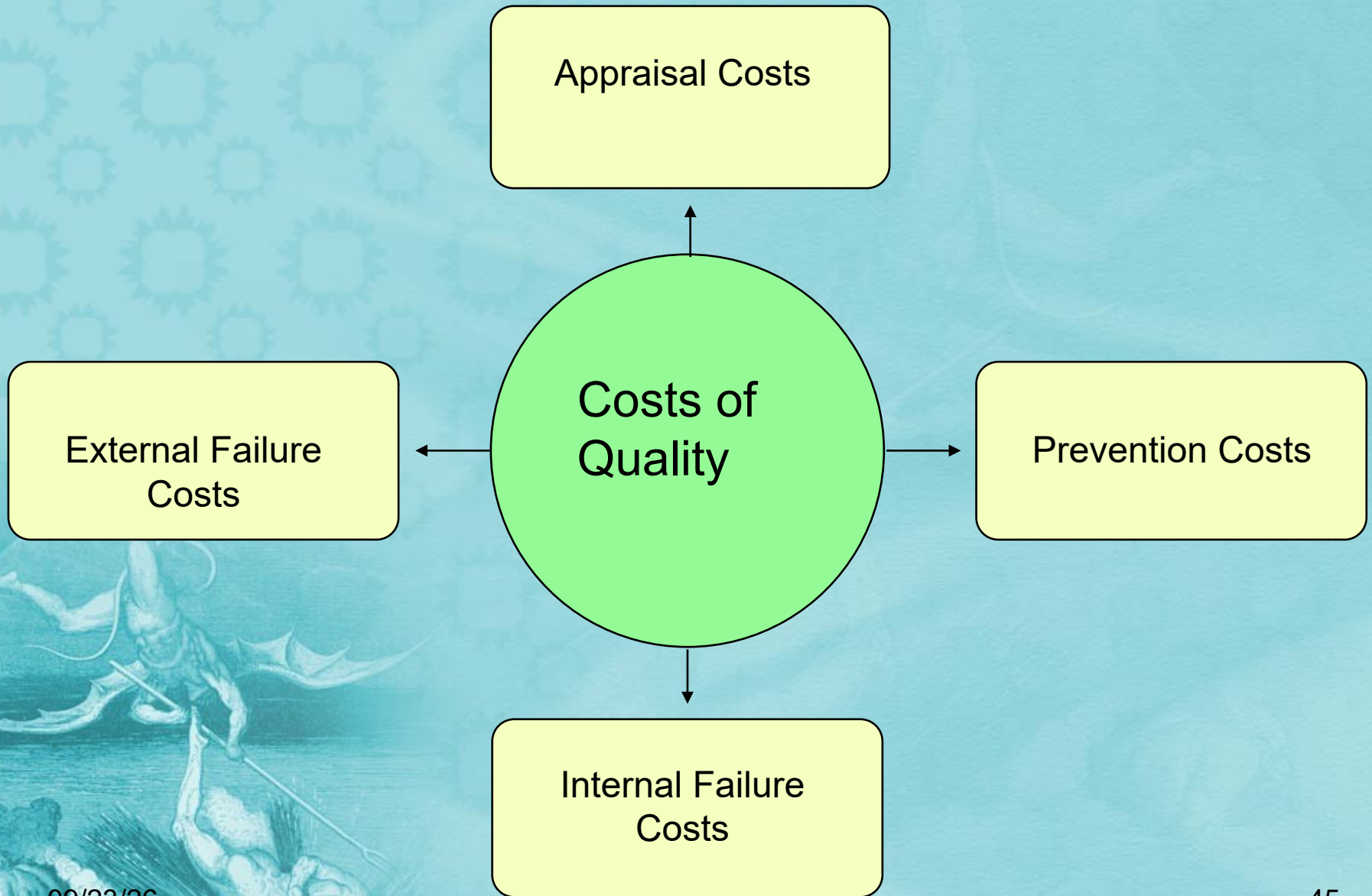


Total Quality Management (TQM)

- **Total quality management** is defined as managing the entire organization so that it excels on all dimensions of products and services that are important to the customer.
- ❑ **TQM** represents a set of management principles that **focus on quality improvement** as the driving force in **all functional areas** and at all levels in a company
- ❑ **Principles of TQM;**
 1. The **customer** define quality, and customer satisfaction is the top priority
 2. Top management must provide the **leadership** for quality

3. Quality is a **strategic issue** and requires a strategic plan
4. Quality is the responsibility of **every employee** at every level of the organization
5. All functions of the company must focus on **continuous quality improvement** to achieve strategic goals
6. Quality problems are solved through **cooperation** among employees and management
7. Problem solving and continuous quality improvement use **statistical quality control** methods
8. **Training and education** of all employees is the basis for continuous quality improvement

Costs of Quality



Cont...

- **Cost of Achieving Good Quality**
 - ✓ **Prevention costs** - costs incurred during product design
 - ✓ **Appraisal costs** - costs of measuring, testing, and analyzing
- **Cost of Poor Quality**
 - ✓ **Internal failure costs** - include scrap, rework, process failure, downtime, and price reductions
 - ✓ **External failure costs** - include complaints, returns, warranty claims, liability, and lost sales

Prevention Costs

- **Quality planning costs**
 - ✓ costs of developing and implementing quality management program
- **Product-design costs**
 - ✓ costs of designing products with quality characteristics
- **Process costs**
 - ✓ costs expended to make sure productive process conforms to quality specifications

- **Training costs**
 - ✓ costs of developing and putting on quality training programs for employees and management
- **Information costs**
 - ✓ costs of acquiring and maintaining data related to quality, and development of reports on quality performance

Appraisal Costs

- **Inspection and testing**
 - ✓ costs of testing and inspecting materials, parts, and product at various stages and at end of process
- **Test equipment costs**
 - ✓ costs of maintaining equipment used in testing quality characteristics of products
- **Operator costs**
 - ✓ costs of time spent by operators to gather data for testing product quality, to make equipment adjustments to maintain quality, and to stop work to assess quality

Internal Failure Costs

- *Scrap costs*
 - ✓ costs of poor-quality products that must be discarded, including labor, material, and indirect costs
- *Rework costs*
 - ✓ costs of fixing defective products to conform to quality specifications
- *Process failure costs*
 - ✓ costs of determining why production process is producing poor-quality products
- *Process downtime costs*
 - costs of shutting down productive process to fix problem
- *Price-downgrading costs*
 - costs of discounting poor-quality products—that is, selling products as “seconds”

External Failure Costs

- **Customer complaint costs**
 - ✓ costs of investigating and satisfactorily responding to a customer complaint resulting from a poor-quality product
- **Product return costs**
 - ✓ costs of handling and replacing poor-quality products returned by customer
- **Warranty claims costs**
 - ✓ costs of complying with product warranties
- **Product liability costs**
 - ✓ litigation costs resulting from product liability and customer injury
- **Lost sales costs**
 - ✓ costs incurred because customers are dissatisfied with poor quality products and do not make additional purchases

Designing Of Quality Control Systems

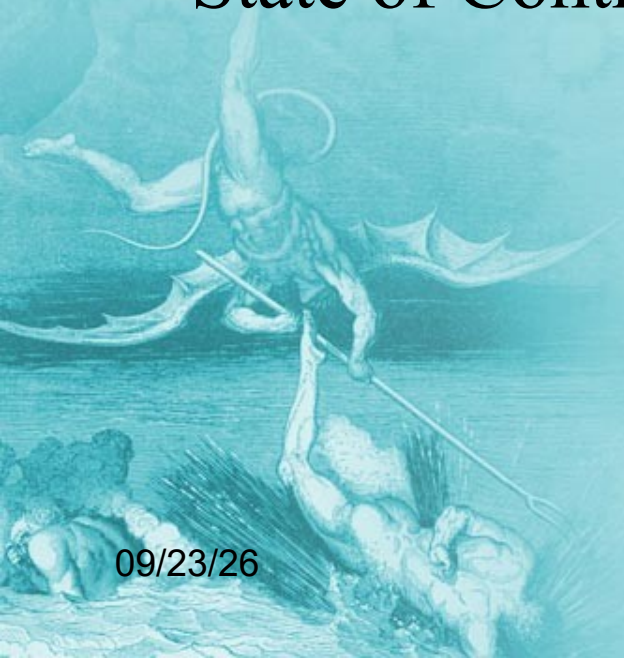
- Processes and “internal customers.”
- “Critical points” and guidelines in identifying them.
- Steps in designing QC systems.

Steps in Designing QC Systems

- ❶ Identify critical points
- ❷ Decide on the type of measurement (variable versus attribute)
- ❸ Decide on the amount of inspection to be used.
- ❹ Decide who should do the inspection

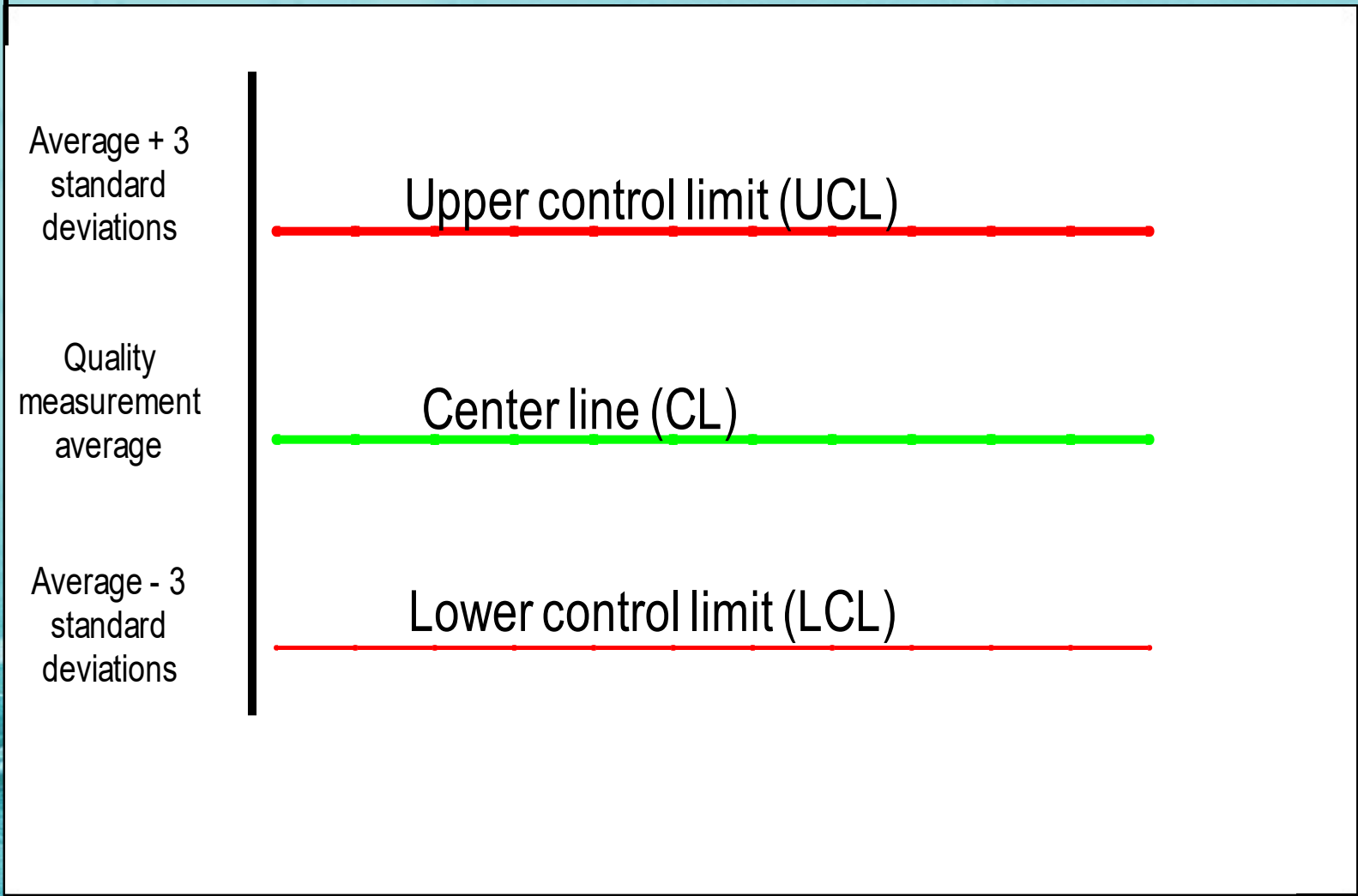
Process Quality Control

- Basic assumptions (tenets) of Process Quality Control:
 - *Every process has random variation in it.*
 - *Production processes are not usually found in a state of control.*
- “State of Control”; what does it mean?

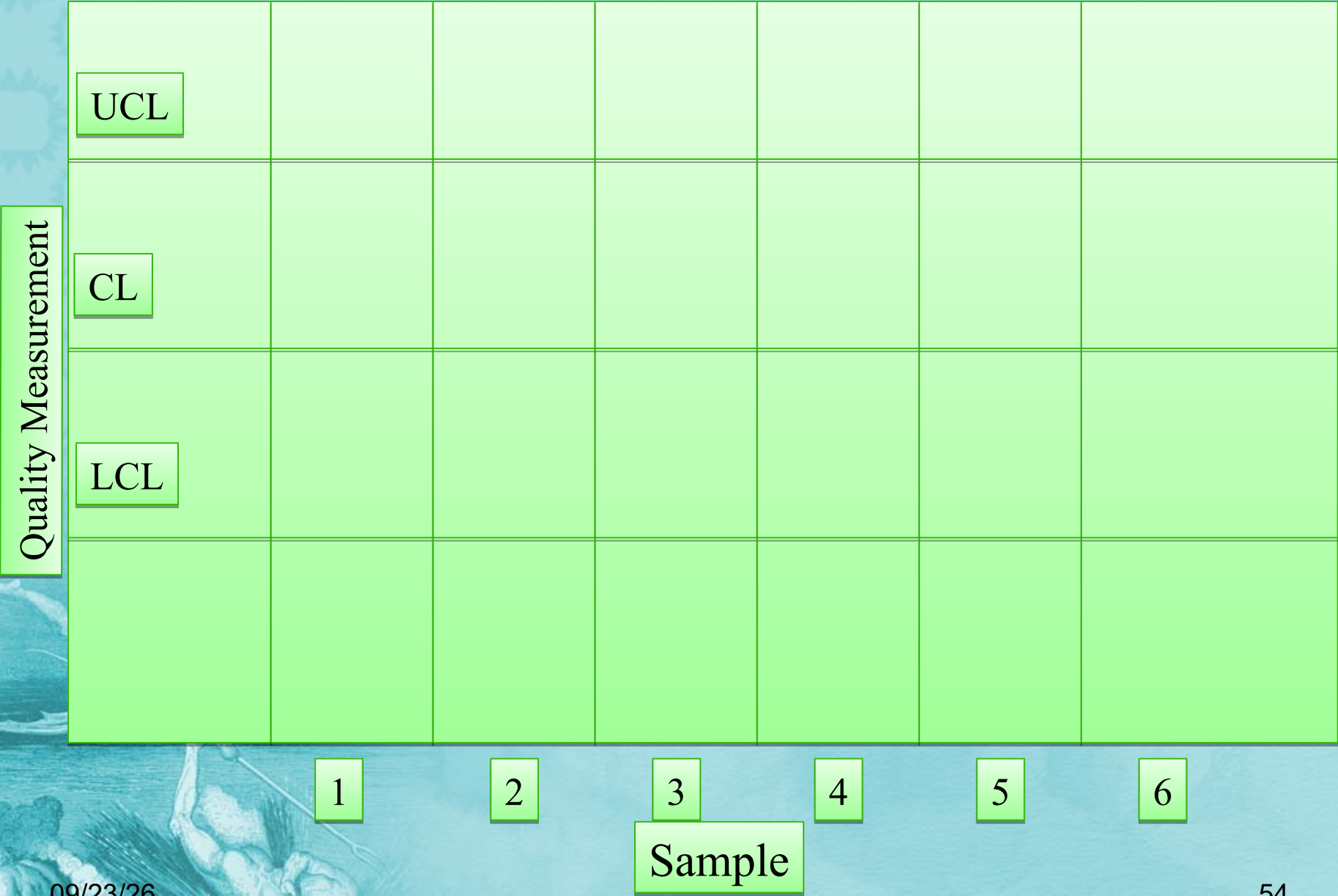


Process Control Chart

Y



Quality Control Chart



Quality Measures: Attributes & Variables

- ❑ **Attribute**: is a product characteristics such as color, surface texture, cleanliness, or smell or taste
- ❑ Hence attributes can be evaluated as: good or bad, acceptable or not, or yes or no
- ❑ **A variable** measure is a product characteristic that is measured on a continuous scale such as length h, temperature, or time

Types of Statistical Sampling

- Attribute (Go or no-go information)
 - ✓ **Defectives** refers to the acceptability of product across a range of characteristics.
 - ✓ **Defects** refers to the number of defects per unit which may be higher than the number of defectives.
 - ✓ **p-chart** application
 - ✓ **C-chart**
- Variable (Continuous)
 - ✓ Usually measured by the mean and the standard deviation.
 - ✓ **X-bar** and **R-chart** applications

Control Limits

- We establish the Upper Control Limits (UCL) and the Lower Control Limits (LCL) with plus or minus 3 standard deviations. Based on this we can expect 99.7% of our sample observations to fall within these limits.



Control Charts for Attributes

- p-charts - uses portion defective in **a sample**
- c-charts - uses number of defects in **an item**

p-Chart

$$UCL = p + z\sigma_p$$

$$LCL = p - z\sigma_p$$

z = number of standard deviations from process average

p = sample proportion defective; an estimate of process average

σ_p = standard deviation of sample proportion

$$\sigma_p = \sqrt{\frac{p(1 - p)}{n}}$$

Cont...

$$p = \frac{\text{Total Number of Defectives}}{\text{Total Number of Observations}}$$

$$s_p = \sqrt{\frac{\bar{p}(1 - \bar{p})}{n}}$$

p-Chart Example

SAMPLE	NUMBER OF DEFECTIVES	PROPORTION DEFECTIVE
1	6	.06
2	0	.00
3	4	.04
:	:	:
:	:	:
20	18	.18
	200	

20 samples of 100 pairs of jeans

Cont...

$$\bar{p} = \frac{\text{total defectives}}{\text{total sample observations}} = 200 / 20(100) = 0.10$$

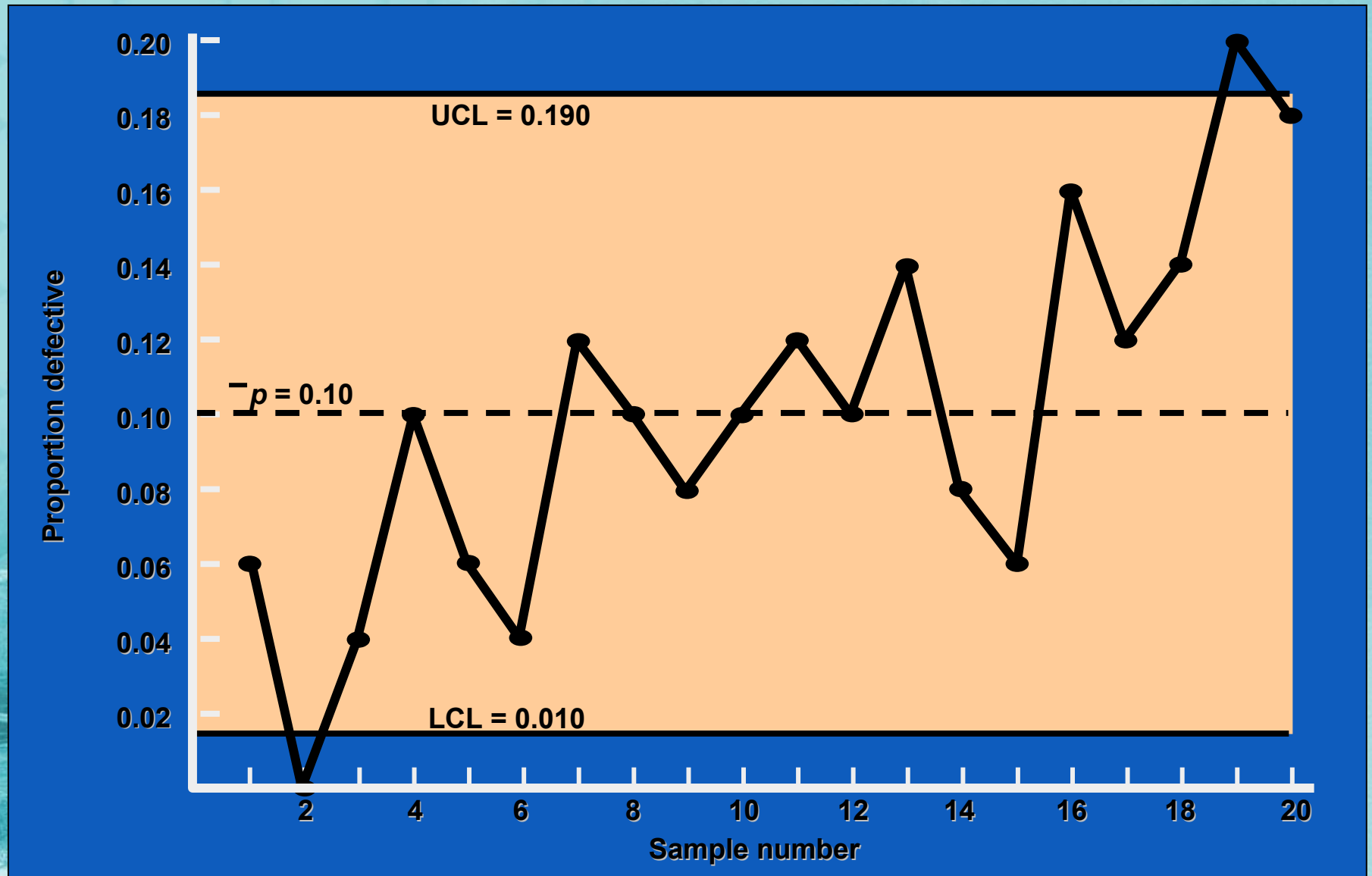
$$UCL = \bar{p} + z\sqrt{\frac{\bar{p}(1 - \bar{p})}{n}} = 0.10 + 3\sqrt{\frac{0.10(1 - 0.10)}{100}}$$

$$UCL = 0.190$$

$$LCL = \bar{p} - z\sqrt{\frac{\bar{p}(1 - \bar{p})}{n}} = 0.10 - 3\sqrt{\frac{0.10(1 - 0.10)}{100}}$$

$$LCL = 0.010$$

Cont...



c-Chart

$$UCL = \bar{c} + z\sigma_c$$

$$LCL = \bar{c} - z\sigma_c$$

$$\sigma_c = \sqrt{\bar{c}}$$

where

c = number of defects per sample

Cont...

Number of defects in 15 sample rooms

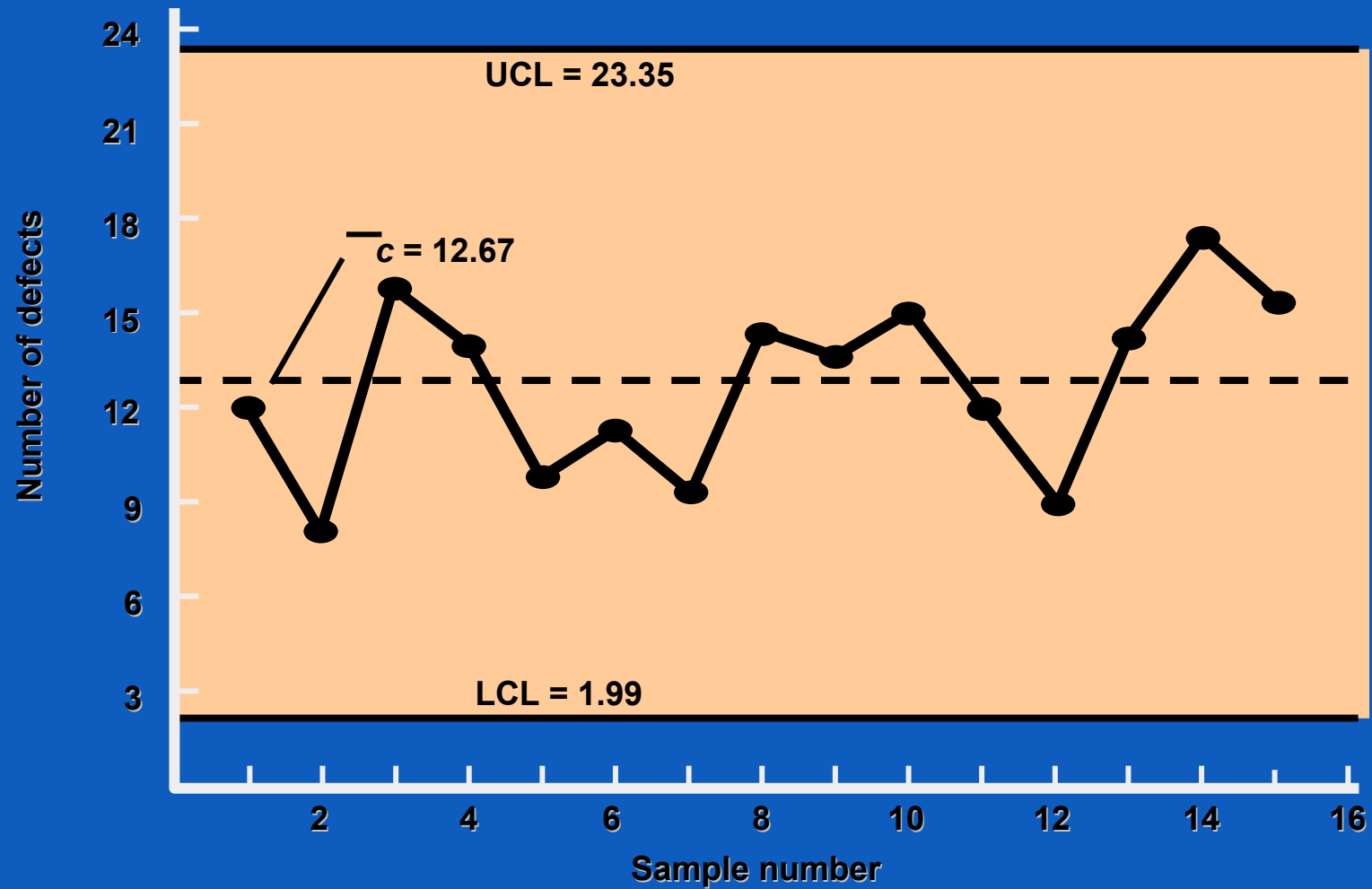
SAMPLE	NUMBER OF DEFECTS
1	12
2	8
3	16
⋮	⋮
⋮	⋮
15	15
	190

$$\bar{c} = \frac{190}{15} = 12.67$$

$$\begin{aligned} \text{UCL} &= \bar{c} + z\sigma_c \\ &= 12.67 + 3 \sqrt{12.67} \\ &= 23.35 \end{aligned}$$

$$\begin{aligned} \text{LCL} &= \bar{c} - z\sigma_c \\ &= 12.67 - 3 \sqrt{12.67} \\ &= 1.99 \end{aligned}$$

Cont...



Control Charts for Variables

- Mean chart ($\bar{x}$ -Chart)
 - ✓ uses average of a sample
- Range chart (R-Chart)
 - ✓ uses amount of dispersion in a sample

$\bar{x}$ -bar Chart

$$\bar{\bar{x}} = \frac{\bar{x}_1 + \bar{x}_2 + \dots + \bar{x}_k}{k}$$

$$UCL = \bar{\bar{x}} + A_2 \bar{R} \quad LCL = \bar{\bar{x}} - A_2 \bar{R}$$

where

=

$\bar{x}$ = average of sample means

x-bar Chart Example

SAMPLE k	OBSERVATIONS (SLIP- RING DIAMETER, CM)					$\bar{x}$	R
	1	2	3	4	5		
1	5.02	5.01	4.94	4.99	4.96	4.98	0.08
2	5.01	5.03	5.07	4.95	4.96	5.00	0.12
3	4.99	5.00	4.93	4.92	4.99	4.97	0.08
4	5.03	4.91	5.01	4.98	4.89	4.96	0.14
5	4.95	4.92	5.03	5.05	5.01	4.99	0.13
6	4.97	5.06	5.06	4.96	5.03	5.01	0.10
7	5.05	5.01	5.10	4.96	4.99	5.02	0.14
8	5.09	5.10	5.00	4.99	5.08	5.05	0.11
9	5.14	5.10	4.99	5.08	5.09	5.08	0.15
10	5.01	4.98	5.08	5.07	4.99	5.03	0.10
						50.09	1.15

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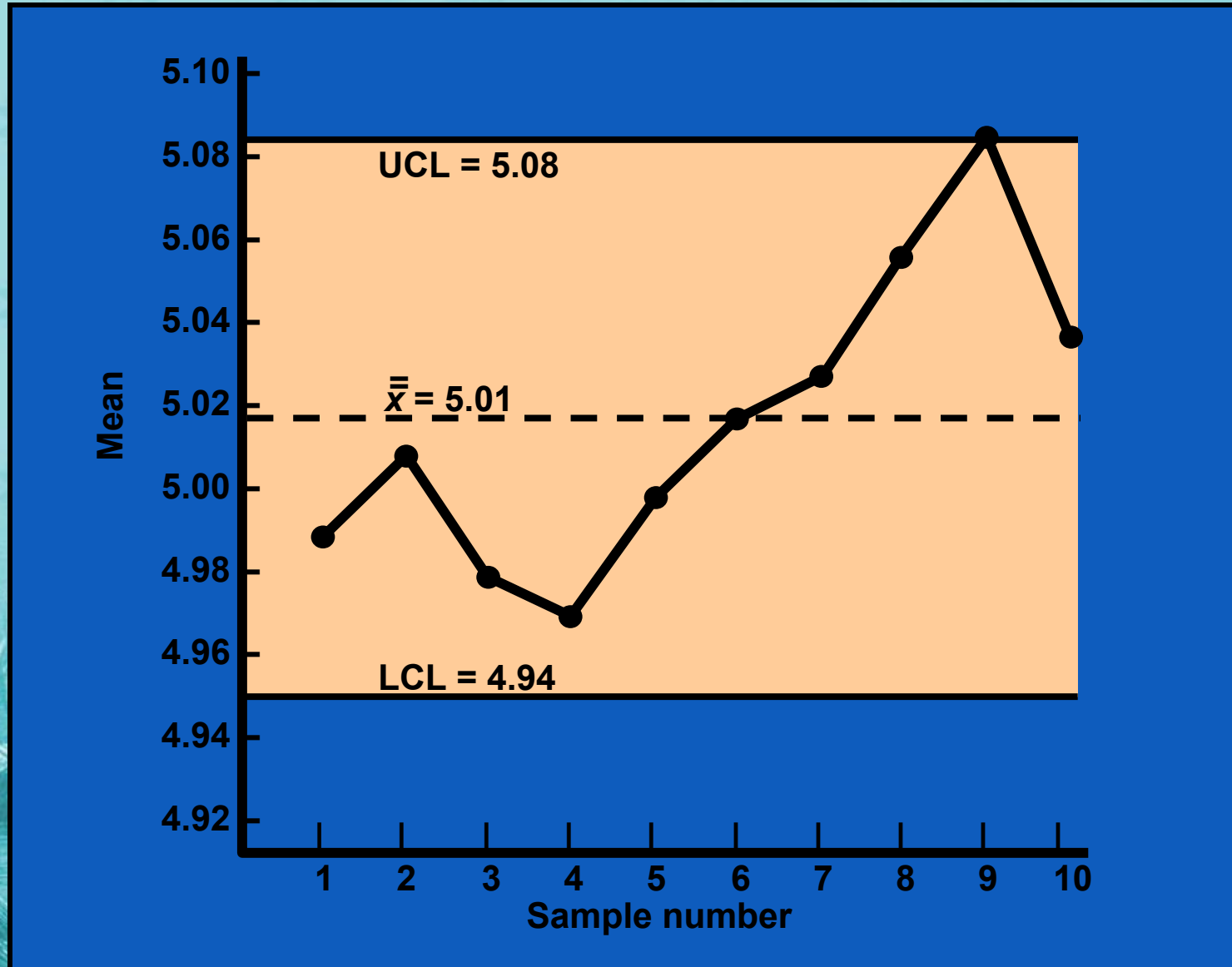
$$\bar{\bar{x}} = \frac{\sum \bar{x}}{k} = \frac{50.09}{10} = 5.01 \text{ cm}$$

$$UCL = \bar{\bar{x}} + A_2 R = 5.01 + (0.58)(0.115) = 5.08$$

$$LCL = \bar{\bar{x}} - A_2 R = 5.01 - (0.58)(0.115) = 4.94$$

Retrieve Factor Value A_2

Cont...



R- Chart

$$UCL = D_4 \bar{R}$$

$$LCL = D_3 \bar{R}$$

$$\bar{R} = \frac{\sum R}{k}$$

where

$\bar{R}$ = range of each sample

k = number of samples

R-Chart Example

SAMPLE k	OBSERVATIONS (SLIP-RING DIAMETER. CM)						$\bar{x}$	R
	1	2	3	4	5			
1	5.02	5.01	4.94	4.99	4.96		4.98	0.08
2	5.01	5.03	5.07	4.95	4.96		5.00	0.12
3	4.99	5.00	4.93	4.92	4.99		4.97	0.08
4	5.03	4.91	5.01	4.98	4.89		4.96	0.14
5	4.95	4.92	5.03	5.05	5.01		4.99	0.13
6	4.97	5.06	5.06	4.96	5.03		5.01	0.10
7	5.05	5.01	5.10	4.96	4.99		5.02	0.14
8	5.09	5.10	5.00	4.99	5.08		5.05	0.11
9	5.14	5.10	4.99	5.08	5.09		5.08	0.15
10	5.01	4.98	5.08	5.07	4.99		<u>5.03</u>	<u>0.10</u>
							50.09	1.15

Cont...

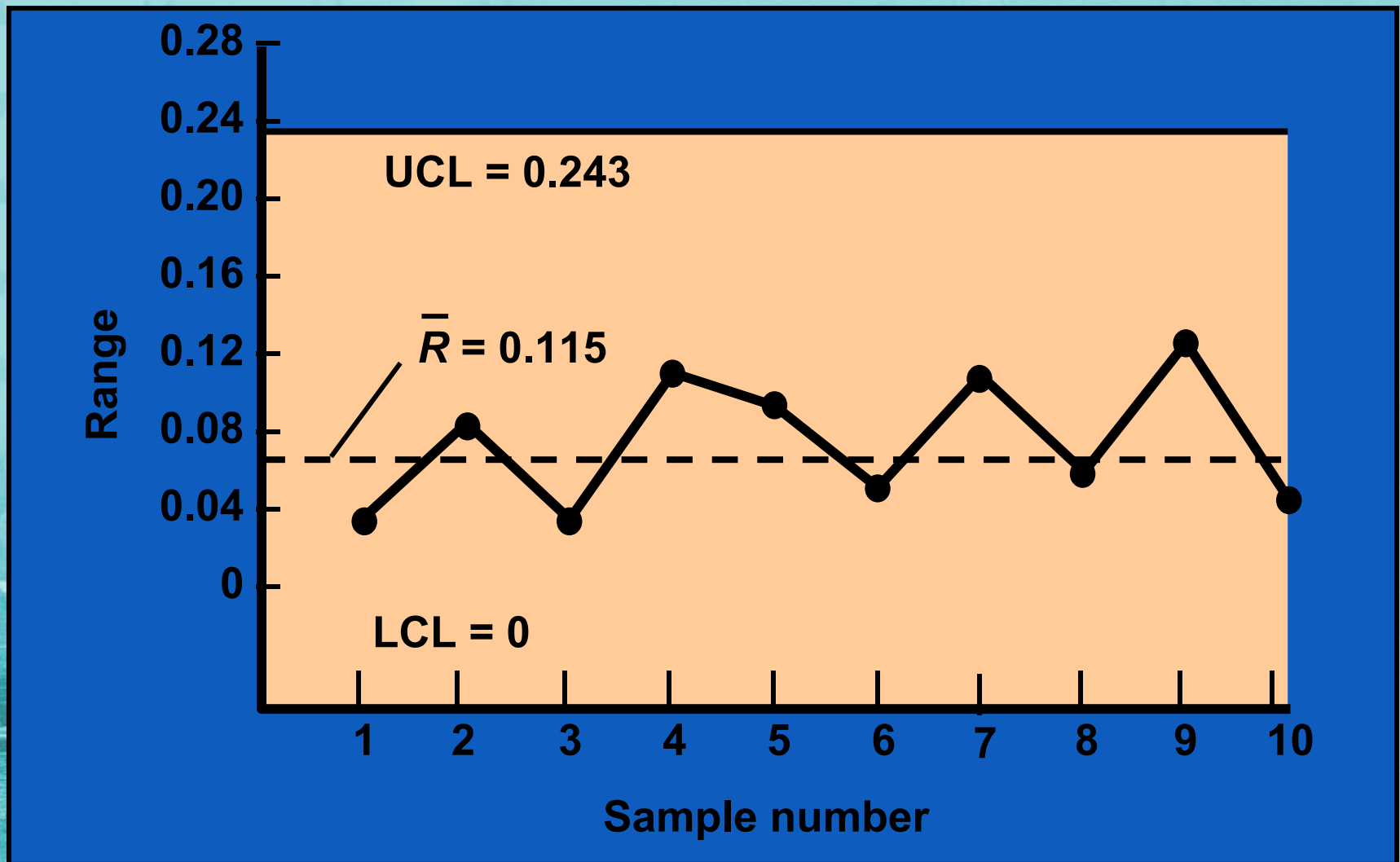
$$\bar{R} = \frac{\sum R}{k} = \frac{1.15}{10} = 0.115$$

$$UCL = D_4 \bar{R} = 2.11(0.115) = 0.243$$

$$LCL = D_3 \bar{R} = 0(0.115) = 0$$

Retrieve Factor Values D_3 and D_4

Cont...



Appendix:

Determining Control Limits for $\bar{x}$ -bar & R-Charts

SAMPLE SIZE n	FACTOR FOR $\bar{x}$ -CHART A_2	FACTORS FOR R-CHART D_3 D_4	
2	1.88	0.00	3.27
3	1.02	0.00	2.57
4	0.73	0.00	2.28
5	0.58	0.00	2.11
6	0.48	0.00	2.00
7	0.42	0.08	1.92
8	0.37	0.14	1.86
9	0.44	0.18	1.82
10	0.11	0.22	1.78
11	0.99	0.26	1.74
12	0.77	0.28	1.72
13	0.55	0.31	1.69
14	0.44	0.33	1.67
15	0.22	0.35	1.65
16	0.11	0.36	1.64
17	0.00	0.38	1.62
18	0.99	0.39	1.61
19	0.99	0.40	1.61
20	0.88	0.41	1.59

Return

Continuous Improvement

- Pareto analysis
- Cause-and-effect (fish-bone) diagrams
- Process capability charts

Check Sheets

Used to keep a record of the number and type of discontinuities over a specified period of time or within a certain batch of product.

Pareto Chart

A graphical representation ranking discontinuities from the most to least significant. Used to help brainstorm what discontinuities, if worked upon first, would be the most likely to produce the greatest improvement in quality.

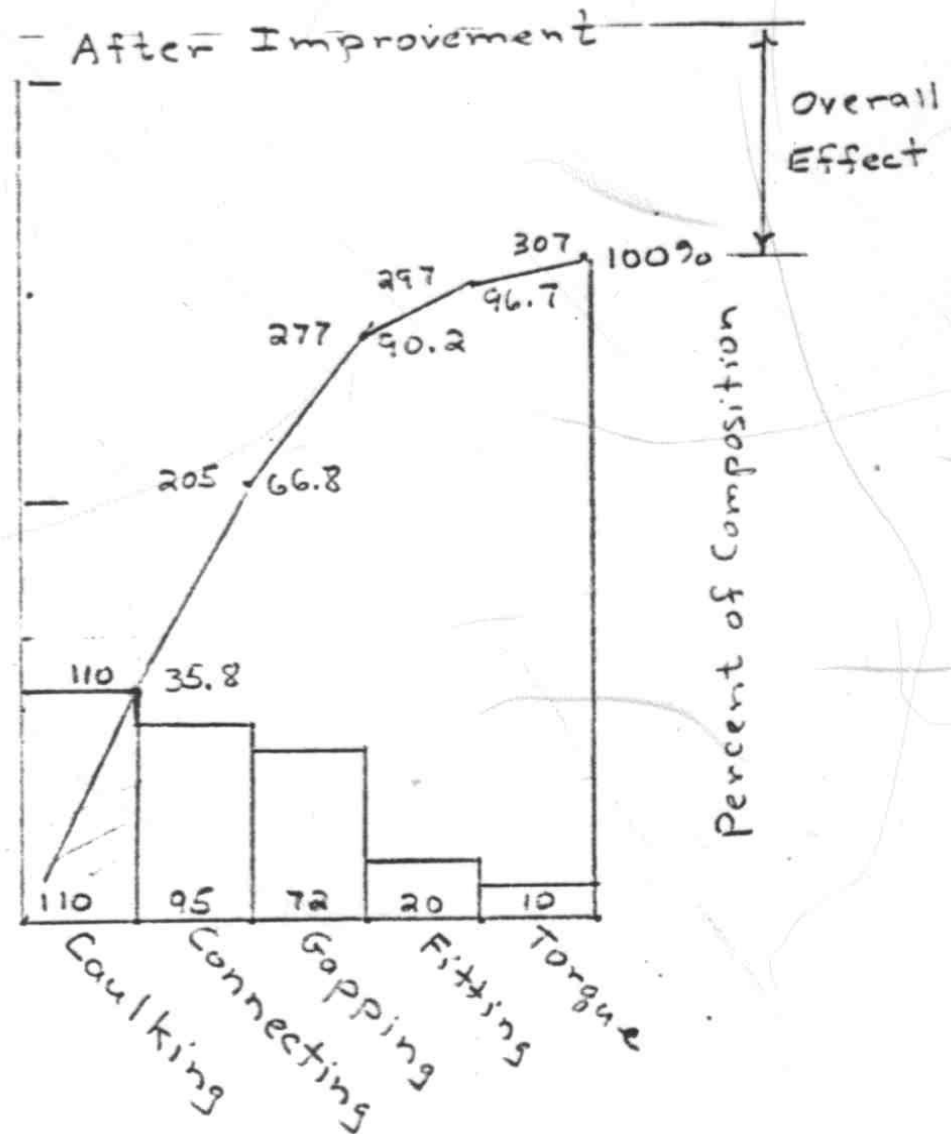
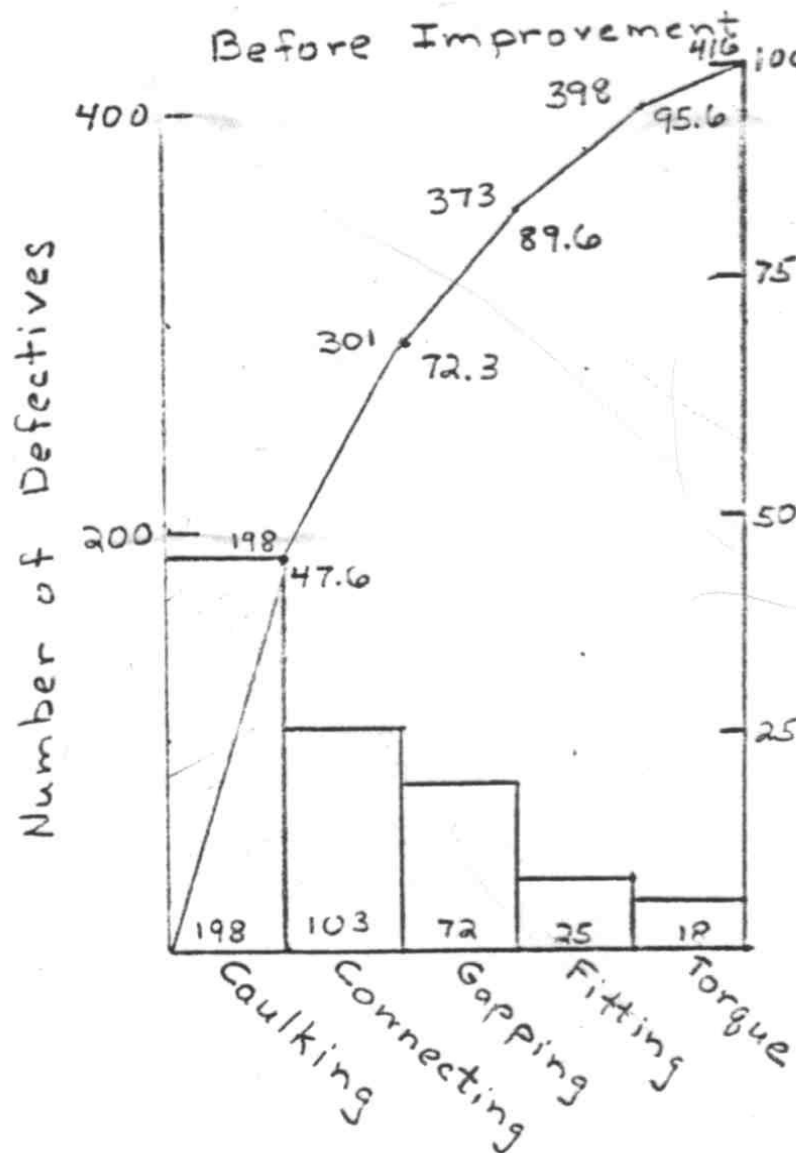
RECORD OF DEFECTIVES

Date: Period since
last analysis:

Number inspected: N=2165

Defective Items	Number of defectives (n)	% of run defective $\frac{n}{N} \times 100$	% of defects per item $\frac{n}{\sum n} \times 100$
Caulking	198	9.1%	47.6%
Fitting	25	1.2%	6.0%
Connections	103	4.8%	24.7%
Torque	18	0.8%	4.3%
Gapping	72	3.3%	17.3%
Total	$\sum n=416$	19.2%	99.9% also called % of composition

Cont...



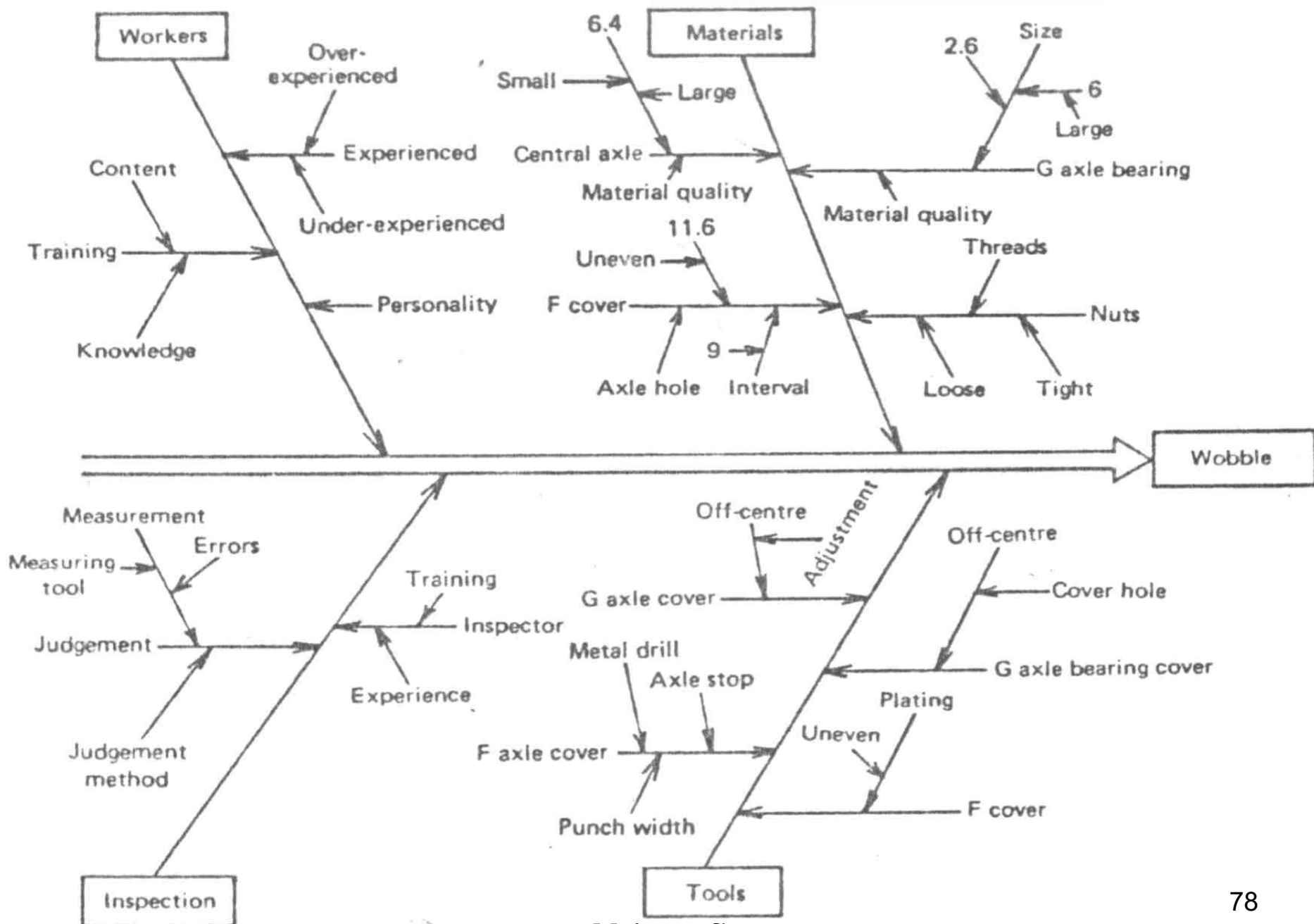
Cause & Effect Diagrams (Fishbone Diagram)

- Used in brainstorming session to help **identify the causes of quality losses**. This diagram is particularly useful after the flow chart and the Pareto diagrams have been developed.



- Step 1: Decide on the quality characteristic {e.g. Reduction of wobble during machine rotation}
- Step 2: Set up the fish bone backbone
- Step 3: Identify main factors causing effect {e.g. Workers, Materials, Inspection, Tools}
- Step 4: Add Cause to each branch

Cont...



Chapter 4

Product and Service Design

Mulugeta G.

**Sheba University College,
Faculty of Business and Economics
Department of Management**

Introduction

- New products & services are the lifeblood of an organization:
 - ✓ It provide a competitive edge by bringing new ideas to the market quickly
 - ✓ Strategically, it defines a firm's customers , as well as its competitors
 - ✓ It capitalizes on a firm's core competencies and determines what new competencies need to be developed

An **effective design** process:

- ✓ Matches product or service characteristics with **customer requirements**
- ✓ Ensures that customer requirements are met in the simplest and **least costly** manner
- ✓ Reduces the **time required** to design a new product or service , and
- ✓ Minimizes the revisions necessary to make a design workable

Product Design

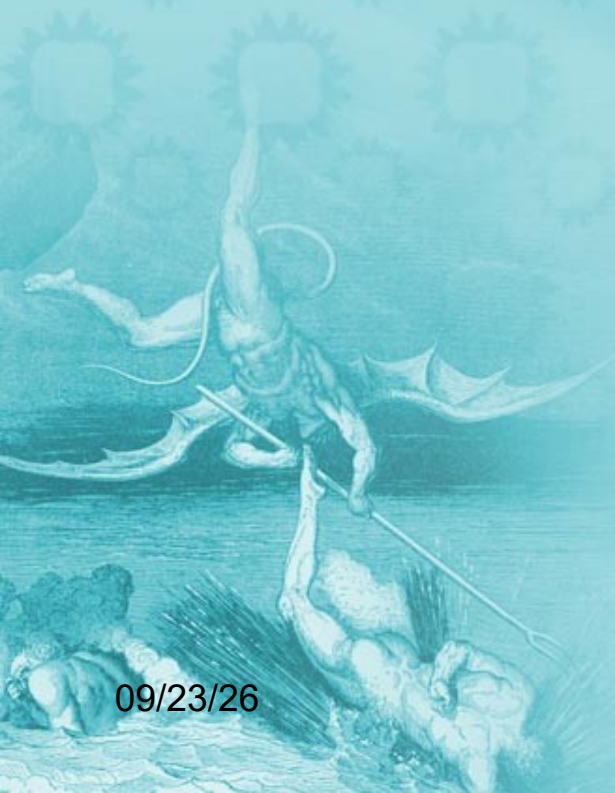
- What is product? What is new product?
- Product design :
 - ✓ Defines the appearance of the product
 - ✓ Sets standards for performance
 - ✓ Specifies which materials are to be used and
 - ✓ Determines dimensions and tolerances

Strategies for New-Product

- Market Pull (*“We Make What We Can Sell”*)
- Technology Push (*“We Sell What We Can Make”*)
- Inter-functional View
- Competitors :
 - ✓ Perceptual maps
 - ✓ Benchmarking and
 - ✓ Reverse engineering

New Product Development Process

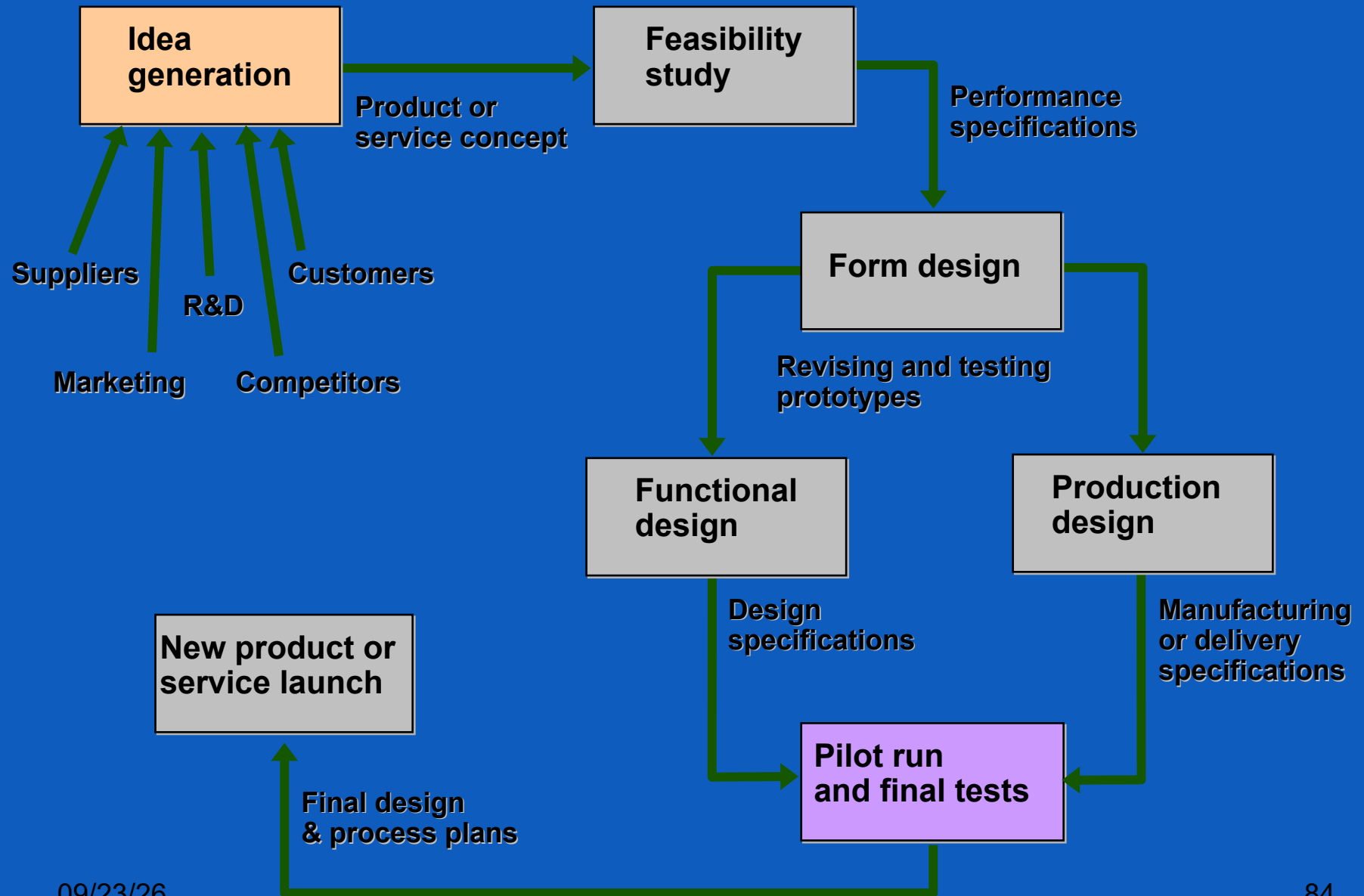
- Concept Development
- Development of Detailed Product Design & Prototyping
- Pilot Production/Testing
- Mass Production and Commercialization



Stage I. Concept Development & Feasibility Study

- This involves: **Generating Ideas for the new product**
- New ideas can be generated from:
 - .. The market
 - .. Within the company (technology push& inter-functional)
 - ..Competitors
- Market surveys , focus groups and interview are important tools to get product ideas from customers.
- Brain storming, panel discussions , Delphi-method etc are important tools in generating ideas from within.
- Marketing and sales departments have big role at this stage .
- The outputs of this stage should be:
 - ..concept design and defining the target market

Design Process



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Idea Generation Sources

- Company's own R&D department
- Customer complaints or suggestions
- Marketing research
- Suppliers
- Salespersons in the field
- Factory workers
- New technological developments
- Competitors

Cont...

- Perceptual Maps
 - ✓ Visual comparison of customer perceptions
- Benchmarking
 - ✓ Comparing product/service against best-in-class
- Reverse engineering
 - ✓ Dismantling competitor's product to improve your own product



Feasibility Study

- At this stage the promising concepts undergo a feasibility study that includes several analyses:
 - ❖ **Market analysis** - assesses whether there is enough demand for the proposed product - this requires customer survey
 - ❖ **Economic analysis** - estimating production and development costs and comparing with estimated sales volume (tools such as cost/benefit analysis , net present value or IRR.
 - ❖ **Technical & strategic analyses** - answer questions such as:
 - ✓ Does the new product require new technology
 - ✓ Is the risk or capital investment excessive
 - ✓ Does the company have sufficient labour and management skills
 - ✓ Does the new product provides a competitive advantage for the company

Stage II. Development of Detailed Product Design and Prototyping

- This stage includes performing the following activities:
 - A. Rapid Prototyping** - This involves building a prototype , testing the prototype, revising the design, retesting etc.
 - B. Form Design**- refers to the physical appearance of product-its shape, colour, size ,and style . Aesthetics such as image , market appeal and personal identification
 - C. Functional Design** - This is concerned with how the product performs . It seeks to meet the performance specifications of fitness for use by the customer

Cont...

- How ever **the detailed product design need to be evaluated in terms** of the following criteria:
 - ❖ Achievement of **customer requirements** and product specifications
 - ❖ Expected **quality and reliability** of the product
 - ❖ Reducibility and **cost** of the product
 - ❖ The impact on production of the company's other products

Stage III: Pilot Production/Testing

This stage requires the following activities :

- ❖ Small quantity production
- ❖ Market test of the sample products
- ❖ Gathering customer opinion on the sample products
- ❖ Based on the customer feedback making the necessary design changes in the product

Stage IV: Mass Production & Commercialization

- Concurrent Design

- A new approach to design that involves simultaneous design of products and processes by design teams

- Improves quality of early design decisions
- Involves suppliers
- Incorporates production process
- Scheduling and management can be complex as tasks are done in parallel

Design For Manufacture & Assembly (DFMA)

- Design for manufacture
 - ✓ design a product for easy and economical production
- Design for assembly
 - ✓ a set of procedures for:
 - reducing number of parts in an assembly
 - evaluating methods of assembly
 - determining an assembly sequence

Design For Manufacture Guidelines

- Minimize number of parts and subassemblies
- Avoid tools, separate fasteners, and adjustments
- Use standard parts when possible and repeatable, well-understood processes
- Design parts for many uses, and modules that can be combined in different ways
- Design for ease of assembly, minimal handling, and proper presentation
- Allow for efficient and adequate testing and replacement of parts

Technology in the Design Process

- Computer Aided Design (CAD)
 - ✓ assists in creation, modification, and analysis of a design; includes:
 - computer-aided engineering (CAE)
 - tests and analyzes designs on computer screen
 - computer-aided manufacturing (CAM)
 - ultimate design-to-manufacture connection

Value analysis (VA)

- Can we do without it?
- Does it do more than is required?
- Does it cost more than it is worth?
- Can something else do a better job?
- Can it be made by
 - ✓ a less costly method?
 - ✓ with less costly tooling?
 - ✓ with less costly material?
- Can it be made cheaper, better, or faster by someone else?

Design for Environment

Design for environment

- ✓ designing a product from material that can be **recycled**
- ✓ design from recycled material
- ✓ design for **ease of repair**
- ✓ minimize packaging
- ✓ minimize material and energy used during manufacture, consumption and disposal

Extended producer responsibility

- ✓ holds companies responsible for their product even after its useful life

Modular Design

- Allow greater variety
- Develop a series of basic product components (modules) for later assembly
- Reduces complexity and costs associated with large number of product variations

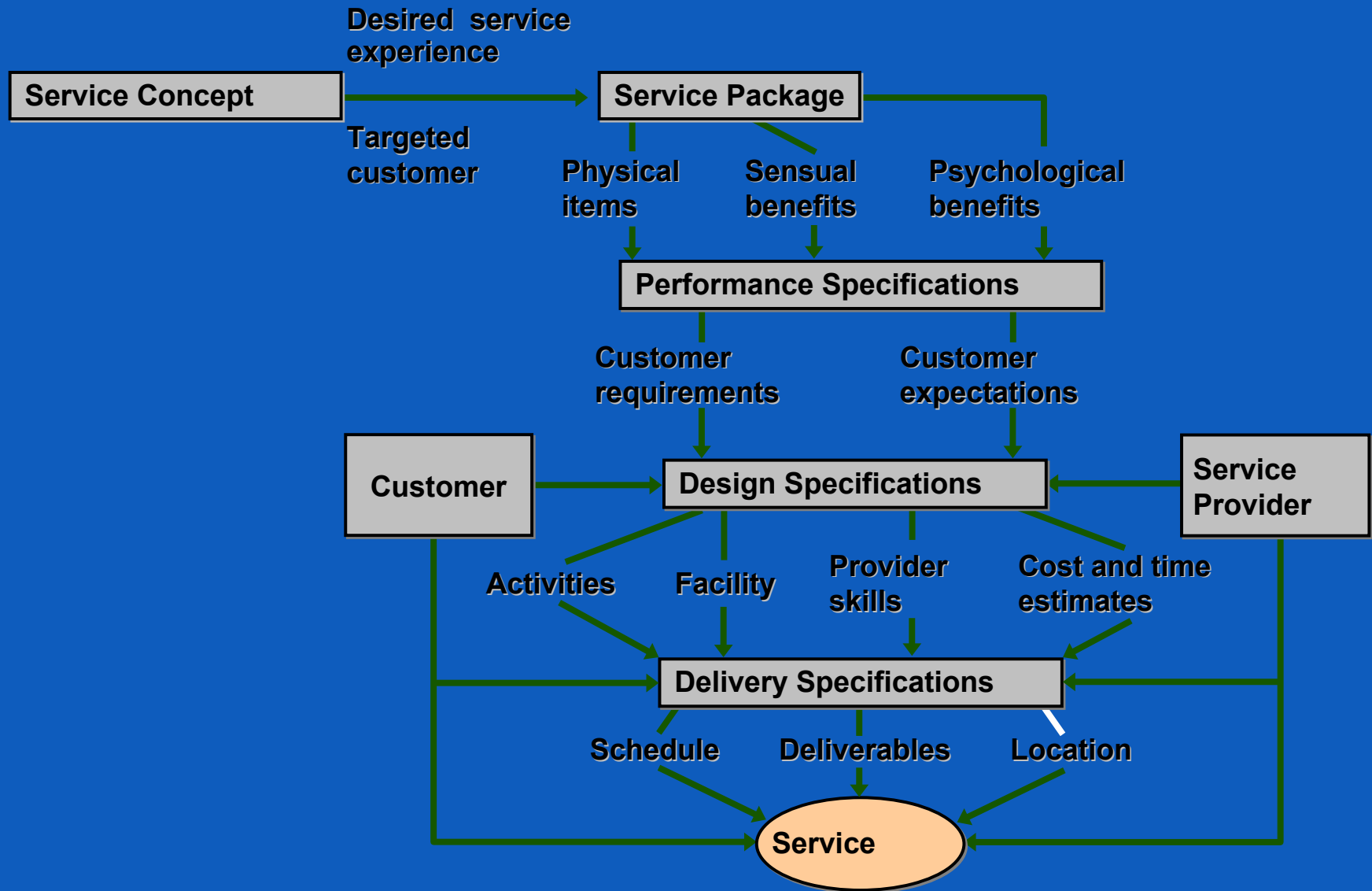
Service Process Design

- **Special Considerations In Service Design**

- Services are intangible
- Service output is variable
- Services have higher customer contact
- Services are perishable

- Service inseparable from delivery
- Services tend to be decentralized and dispersed
- Services are consumed more often than products
- Services can be easily emulated (copied or imitated)

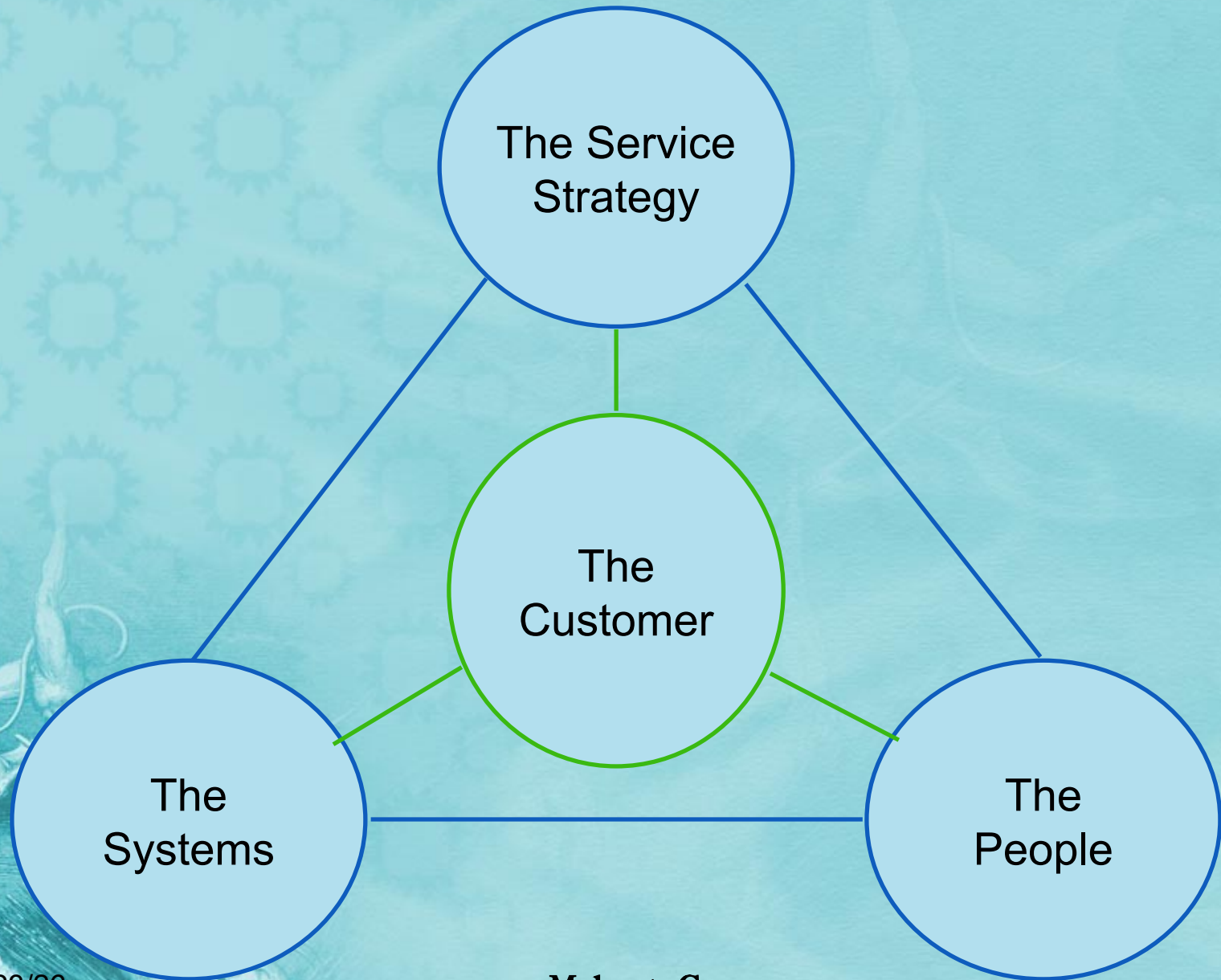
Cont...



Service Design Process

- Service concept
 - ✓ purpose of a service; it defines target market and customer experience
- Service package
 - ✓ mixture of physical items, sensual benefits, and psychological benefits
- Service specifications
 - ✓ performance specifications
 - ✓ design specifications
 - ✓ delivery specifications

The Service Triangle



Service Strategy: Focus And Advantage Performance Priorities

- ✓ Treatment of the customer
- ✓ Speed and convenience of service delivery
- ✓ Price
- ✓ Variety
- ✓ Quality of the tangible goods
- ✓ Unique skills that constitute the service offering

CHAPTER 6

PROCESS SELECTION

Mulugeta G.

Sheba University College
Faculty of Business and Economics
Department of Management

Process Planning

- **Process** - a group of related tasks with specific inputs and outputs
- **Process design** - tasks need to be done and coordinated among functions, people, and organizations
- **Process planning** - converts designs into workable instructions for manufacture or delivery
- **Process strategy** - an organization's overall approach for physically producing goods and services

Process Strategy

- **Capital intensity** - mix of capital (i.e., equipment, automation) and labor resources used in production process
- **Process flexibility** - ease with which resources can be adjusted in response to changes in demand, technology, products or services, and resource availability
- **Vertical integration** - extent to which firm will produce inputs and control outputs of each stage of production process
- **Customer involvement** - role of customer in production process

What is Process Selection?

- **Process Selection** refers to the strategic decision of selecting which kind of production processes to have in the plant .

Classification of Processes:

- ✓ Based on what they do:
 - **Conversion processes** like changing iron ore into steel
 - **Fabrication processes** like changing raw materials into some specific form e.g. changing steel into spare parts
 - **Assembly processes** like assembling a fender into a car (assembling parts to form a car)
- ✓ Based on how they are designed:
 - Make-to-stock
 - Make-to-order

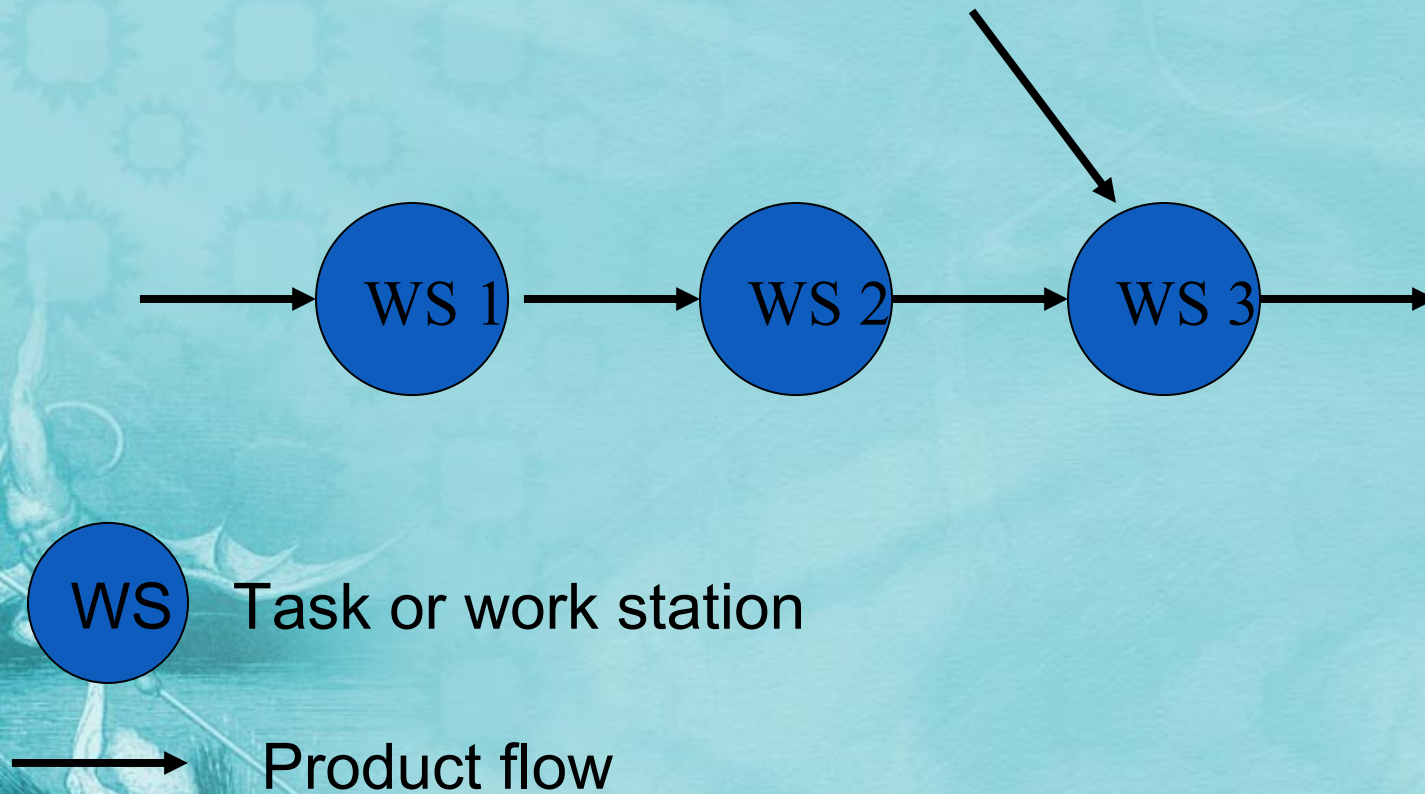
Product-Flow Characteristics

Types of Product Flow

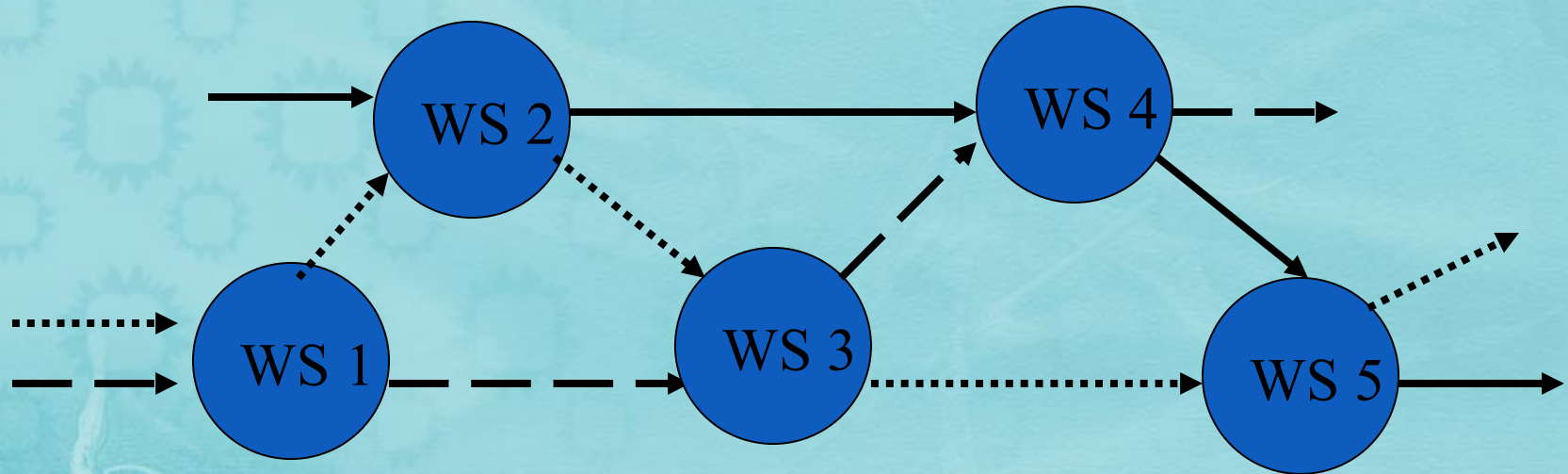
- Line Flow
 - ✓ Repetitive/Assembly
 - ✓ Continuous
- Intermittent Flow
 - ✓ Batch flow
 - ✓ Job shop flow
- Project Flow

Characteristics of Flows

Line Flow



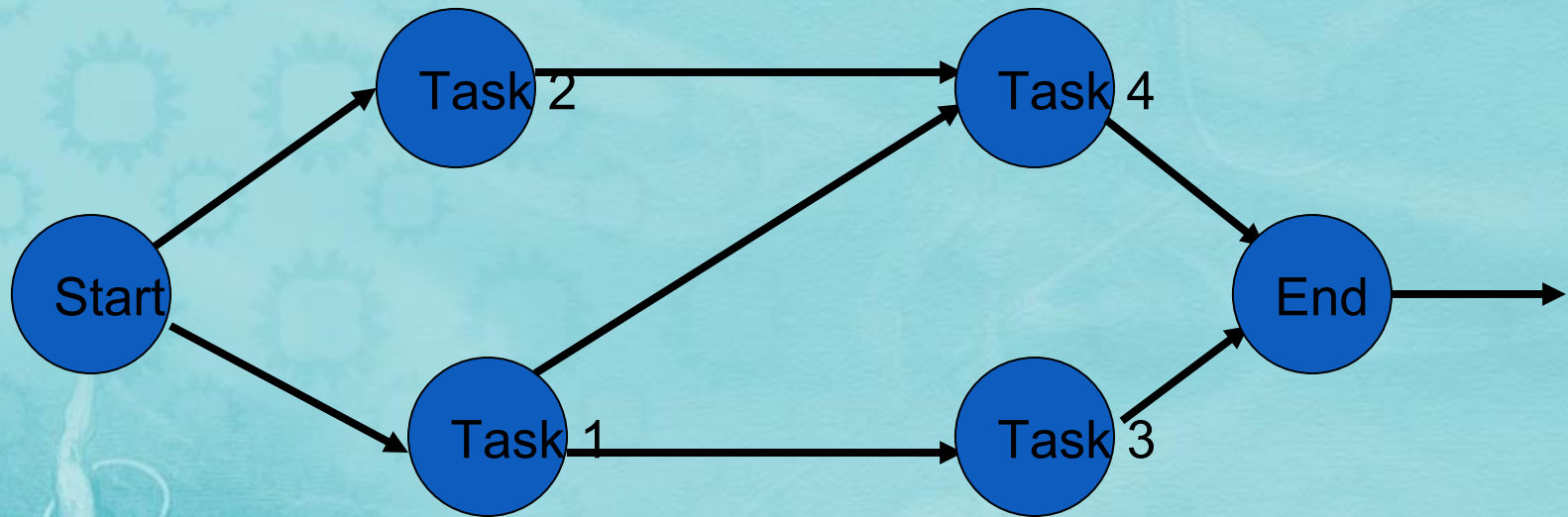
Intermittent Flow



WS Task or work station

— — — — — Product flows
- - - - -
— — — — —

Project Flow



Task Task or activity

→ Precedence relationship

Process Types

- **Job shop**
 - ✓ Small scale
- **Batch**
 - ✓ Moderate volume
 - ✓ systems process many different jobs through the system in groups or batches
- **Repetitive/assembly line**
 - ✓ High volumes of standardized goods or services
- **Continuous**
 - ✓ Very high volumes of non-discrete goods
 - ✓ used for very-high volume commodity products

Types of Processes

	PROJECT	BATCH	MASS	CONTINUOUS
Type of product	Unique	Made-to-order (customized)	Made-to-stock (standardized)	Commodity
Type of customer	One-at-a-time	Few individual customers	Mass market	Mass market
Product demand	Infrequent	Fluctuates	Stable	Very stable

Source: Adapted from R. Chase, N. Aquilano, and R. Jacobs, *Operations Management for Competitive Advantage* (New York:McGraw-Hill, 2001), p. 210

Cont...

	Job Shop	Batch	Repetitive/ Assembly	Continuous
Description	Customized goods or services	Semi-standardized goods or services	Standardized goods or services	Highly standardized goods or services
Advantages	Able to handle a wide variety of work	Flexibility	Low unit cost, high volume, efficient	Very efficient, very high volume
Disadvantages	Slow, high cost per unit, complex planning and scheduling	Moderate cost per unit, moderate scheduling complexity	Low flexibility, high cost of downtime	Very rigid, lack of variety, costly to change, very high cost of down time
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Cont...

Variety

Job shop

Emergency
Room

Batch

Commerci
al Bakery

Repetitive

Automatic
Carwash

Continuou
s

Petroleum
Refinery

Low



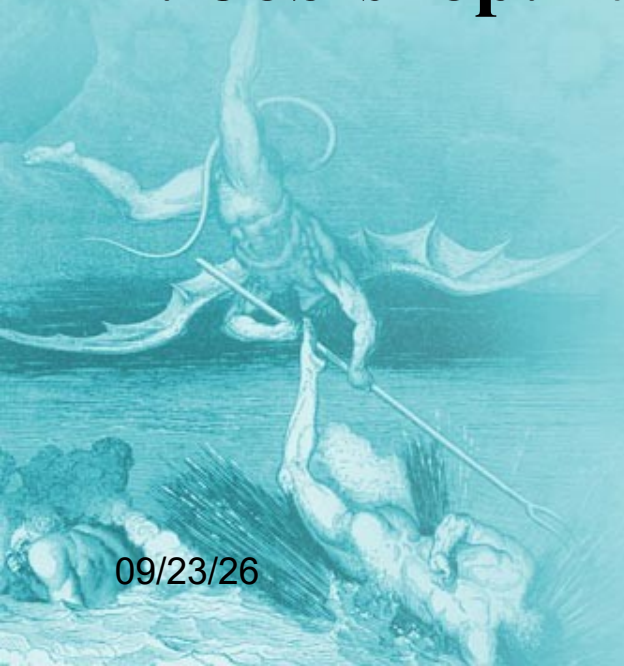
Volume



High

Examples on the process type

- A. Continuous flow:** paper, chemicals, petroleum, metals and water
- B. Repetitive or assembly or discrete flow:** computers, automobiles, cafeteria-style meals
- C. Batch process:** books, toys, canned vegetables
- D. Job-shop:** hair style, courses, furniture, retail-sales



Factors Affecting Process Choice

- Market conditions and competition
- Capital requirements
- Labor supply and cost
- Management skills
- Materials supply and cost
- State of technology
- The expected **volume** and **demand pattern** for the products
- The number of different products (**product variety**) to be made by the system and the types of processing each requires
- The customer **order type**: make-to-stock and make-to-order
- The physical characteristics of the products

CHAPTER 6

Facility Layout

Mulugeta G.

Sheba University College
Faculty of Business and Economics
Department of Management

Facility layout

- Facility layout refers to the format in which departments, workgroups within departments, workstations, machines and stock-holding points within a production facility are arranged.
- The aim of facility layout is **smooth flow of work**, materials and customers through the production system.

Importance of Facility Layout Decisions

Facility layout decisions are important in that:

- ✓ They require substantial investments of money and effort
- ✓ They involve long-term commitments, which make mistakes difficult to overcome, and
- ✓ They have a significant impact on the cost and efficiency of short-term operations.

Objectives of Facility Layout Decisions

- Minimize material handling costs
- Utilize space efficiently
- Utilize labor efficiently
- Eliminate bottlenecks
- Facilitate communication and interaction between workers, between workers and their supervisors, or between workers and customers
- Reduce manufacturing cycle time or customer service time
- Eliminate waste or redundant movement
- Facilitate the entry, exit, and placement of material, products, or people
- Incorporate safety and security measures
- **Promote product and service quality**
- Provide a visual control of operations or activities guarding premises be located?
- Provide flexibility to adapt to changing conditions

Characteristics of the Facility Layout Decisions

- Location of the various working areas impacts the flow through the system.
- The layout can affect productivity and costs generated by the system.
- Layout alternatives are limited by:
 - ✓ **amount and type of space required** for the various areas
 - ✓ **amount and type of space available**
 - ✓ **operations strategy**
- Layout decisions tend to:
 - ✓ **Be infrequent**
 - ✓ **Be expensive to implement**
 - ✓ **Be studied and evaluated extensively**
 - ✓ **Involve long-term commitment or resources**

CHAPTER 7

Facility Location

Mulugeta G.

Sheba University College
Faculty of Business and Economics
Department of Management

Introduction: Definition

- **Facility location** is the process of identifying the best geographic location for a service or production facility.
- Location decisions are important in that:
 - They affect firms' sales potential
 - Involve long term commitment of resources.
 - Affect firms' operating costs

Issues in Facility Location

- Proximity to Customers
- Business Climate
- Total Costs
- Infrastructure
- Quality of Labor
- Suppliers
- Other Facilities
- Political Risk
- Government Barriers
- Trading Blocs
- Environmental Regulation
- Competitive Advantage

Facility Location Decisions

The Need for Facility Location Decisions

1. For existing businesses, location decisions are tools for marketing strategies.
 - ✓ In such cases, location decisions refer to additional locations not new ones.
2. Inability of existing businesses to meet growth in demand by making expansions.
3. Depletion of basic raw materials in one location.
4. Shift in markets for existing products

The Nature of Location Decisions

- a. Importance of Location Decisions
- b. Objectives of Locations and
- c. Location options

Importance of Location Decisions

- Location decisions are important in that:
 - i. They involve long-term commitment of resources
 - ii. They determine investment and operating costs
 - iii. They determine revenue or sales volume
 - iv. Availability of qualified labor
 - v. Availability of suppliers

Objectives of Location Decisions

- Sales generation
- Cost Minimization
- Profit maximization
- Accessing raw materials
- Proximity to customers and delivering services to all customers

❖ **In any situation, there can never be a single best location.**

Location Options

- Three Location options are generally considered in making location decisions.

1. Expansion

- Realistic only if there exists adequate space.
- Preferable only if the existing location has special features that other locations do not have.

2. Adding a new location

- This is important for:
 - a. Counter attacks against rivals
 - b. Expanding markets in new locations
 - c. Defensive strategy against new entrants

3. Moving into a New Location

The causes for moving into a new location can be either:

- a. A shift in markets or
- b. exhaustion of raw materials or
- c. the cost of operations

Procedures for Making Location Decisions

1. Decide on the criteria to use for evaluating location alternatives, such as increased revenues or community service.
2. Identify important factors, such as location of markets or raw materials.
3. Develop location alternatives:
 - a. Identify the general region for a location
 - b. Identify a small number of community alternatives
 - c. Identify site alternatives among the community alternatives.
4. Evaluate the alternatives and make a selection.

The Location Decision Stages and factors Affecting Facility Location

- Facility location decisions are commonly made in five stages:
 - ✓ The Regional Decision
 - ✓ The Local Decision
 - ✓ The Site Decision
 - ✓ Community related factor and
 - ✓ Labour related factor

The Regional Decision

- A region may be a **country**, **part of a country** or **province** (region or state)
- At this stage: **economic, market and legal** factors are dominant
- The Following are specific factors of potential importance:
 - ✓ Market proximity
 - ✓ Proximity to raw materials
 - ✓ Availability of utilities
 - ✓ labour supply and unionization and

Additional factors for international location decision:

- ✓ National taxes –profit taxes Vs value added taxes
- ✓ Legal restrictions

The Local Decision

- This involves selecting among cities , metropolitan areas etc.
- For example a company may decide to locate within Zone One of Afar Region. Within this zone, the possible local alternatives might be :Samara, Logia, Chifra, Dubti etc.
- At this point , the following additional location factors are relevant for consideration:
 1. Taxes
 2. Economic incentives - free land, low-cost loans or tax abatements , employee training, leasing cost
 3. Attractiveness of the community
 4. Quality of housing, rate of crime , quality of schools, recreational areas, etc.
 5. Compatible Industry and Transportation Network
 6. Government policy and Attitude
 7. Environmental Regulations

Global Location Factors

- Government stability
- Government regulations
- Political and economic systems
- Economic stability and growth
- Exchange rates
- Culture
- Climate
- Export import regulations, duties and tariffs
- Raw material availability
- Number and proximity of suppliers
- Transportation and distribution system
- Labor cost and education
- Available technology
- Commercial travel
- Technical expertise
- Cross-border trade regulations
- Group trade agreements

Regional Location Factors (Summary)

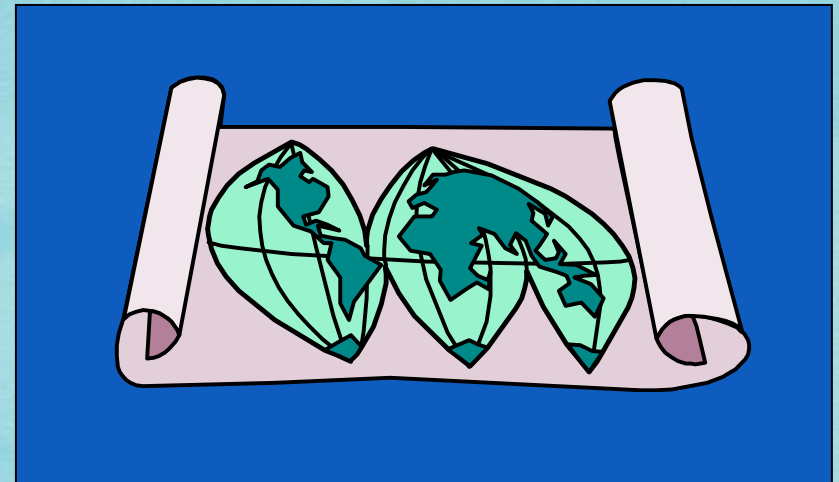
- Labor (availability, education, cost, and unions)
- Proximity of customers
- Number of customers
- Construction/leasing costs
- Land cost
- Business climate
- Community services
- Incentive packages
- Government regulations
- Environmental regulations
- Raw material availability
- Commercial travel & Climate
- Modes and quality of transportation
- Transportation costs
- Community government
Local business regulations
- Government services (e.g., Chamber of Commerce)
- Infrastructure (e.g., roads, water, sewers)
- Quality of life
- Taxes
- Availability of sites
- Financial services
- Community inducements
- Proximity of suppliers
- Education system

Location Incentives

- Tax credits
- Relaxed government regulation
- Job training
- Infrastructure improvement
- Money

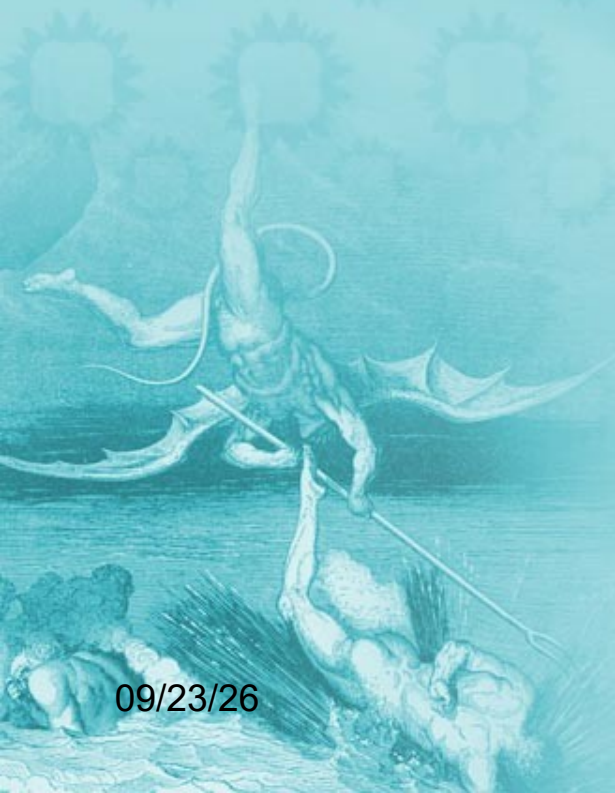
Location Analysis Techniques

- Location rating factor
- Center-of-gravity
- Load-distance



Location Rating Factor

- Identify important factors
- Weight factors (0.00 - 1.00)
- Subjectively score each factor (0 - 100)
- Sum weighted scores



Location Factor Rating: Example

SCORES (0 TO 100)				
LOCATION FACTOR	WEIGHT	<i>Mekelle</i>	<i>Wukro</i>	<i>Addis</i>
Labor pool and climate	.30	80	65	90
Proximity to suppliers	.20	100	91	75
Wage rates	.15	60	95	72
Community environment	.15	75	80	80
Proximity to customers	.10	65	90	95
Shipping modes	.05	85	92	65
Air service	.05	50	65	90
Weighted Score for “Labor pool and climate” for Mekelle =weighted value x factor score=(0.30)(80) = 24				

Cont...

WEIGHTED SCORES

<i>Mekelle</i>	<i>Wukro</i>	<i>Addis</i>
24.00	19.50	27.00
20.00	18.20	15.00
9.00	14.25	10.80
11.25	12.00	12.00
6.50	9.00	9.50
4.25	4.60	3.25
2.50	3.25	4.50
77.50	80.80	82.05

Addis has the
highest factor rating

Center-of-Gravity Technique

- ✓ Center of gravity method - is a technique for locating single facilities that **considers the existing facilities, the distances between them**, and the volumes of goods to be shipped.
- ✓ It is a method to **determine the location of a distribution center that will minimize distribution costs**.
- ✓ It treats distribution cost as a linear function of the distance and the quantity shipped.
- ✓ quantity to be shipped to each destination is assumed to be fixed (i.e., will not change over time).
- ✓ Locate facility at center of geographic area
- ✓ Based on weight and distance traveled establish grid-map of area
- ✓ Identify coordinates and weights shipped for each location

Center of Gravity Method Formulas

$$C_x = \frac{\sum d_{ix} V_i}{\sum V_i} \quad C_y = \frac{\sum d_{iy} V_i}{\sum V_i}$$

C_x = X coordinate of center of gravity

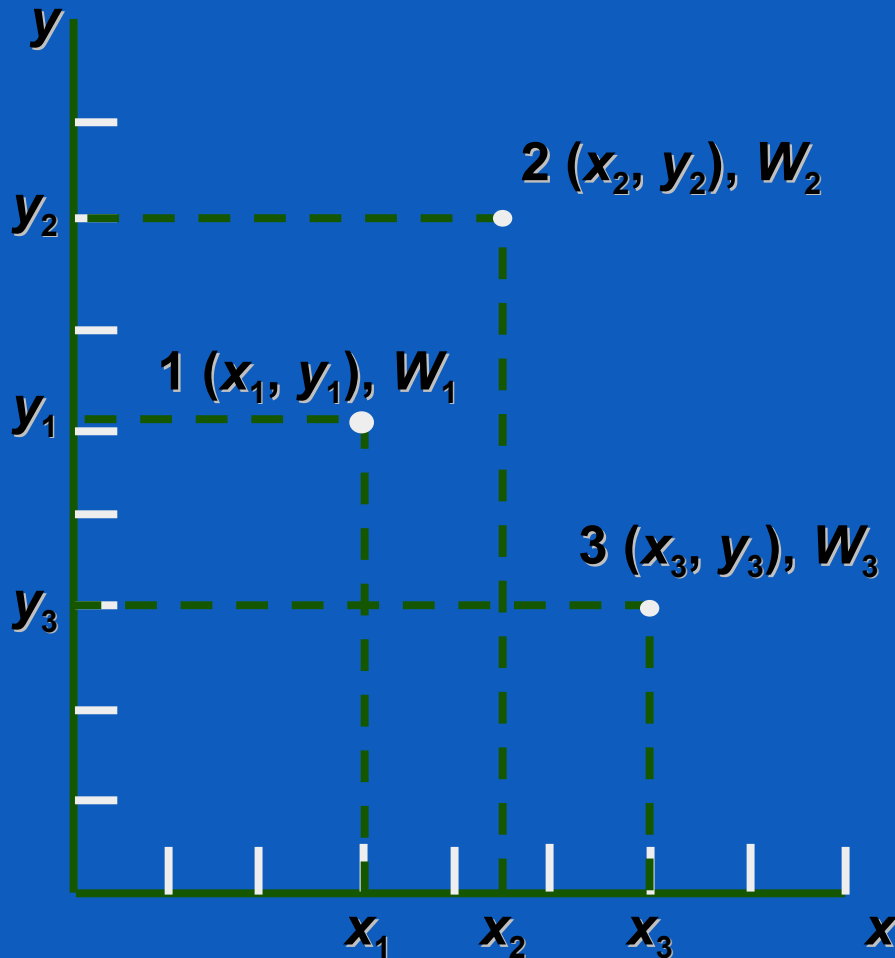
C_y = Y coordinate of center of gravity

d_{ix} = X coordinate of the i^{th} location

d_{iy} = Y coordinate of the i^{th} location

V_i = volume of goods moved to or from i^{th} location

Grid-Map Coordinates



$$x = \frac{\sum_{i=1}^n x_i W_i}{\sum_{i=1}^n W_i} \quad y = \frac{\sum_{i=1}^n y_i W_i}{\sum_{i=1}^n W_i}$$

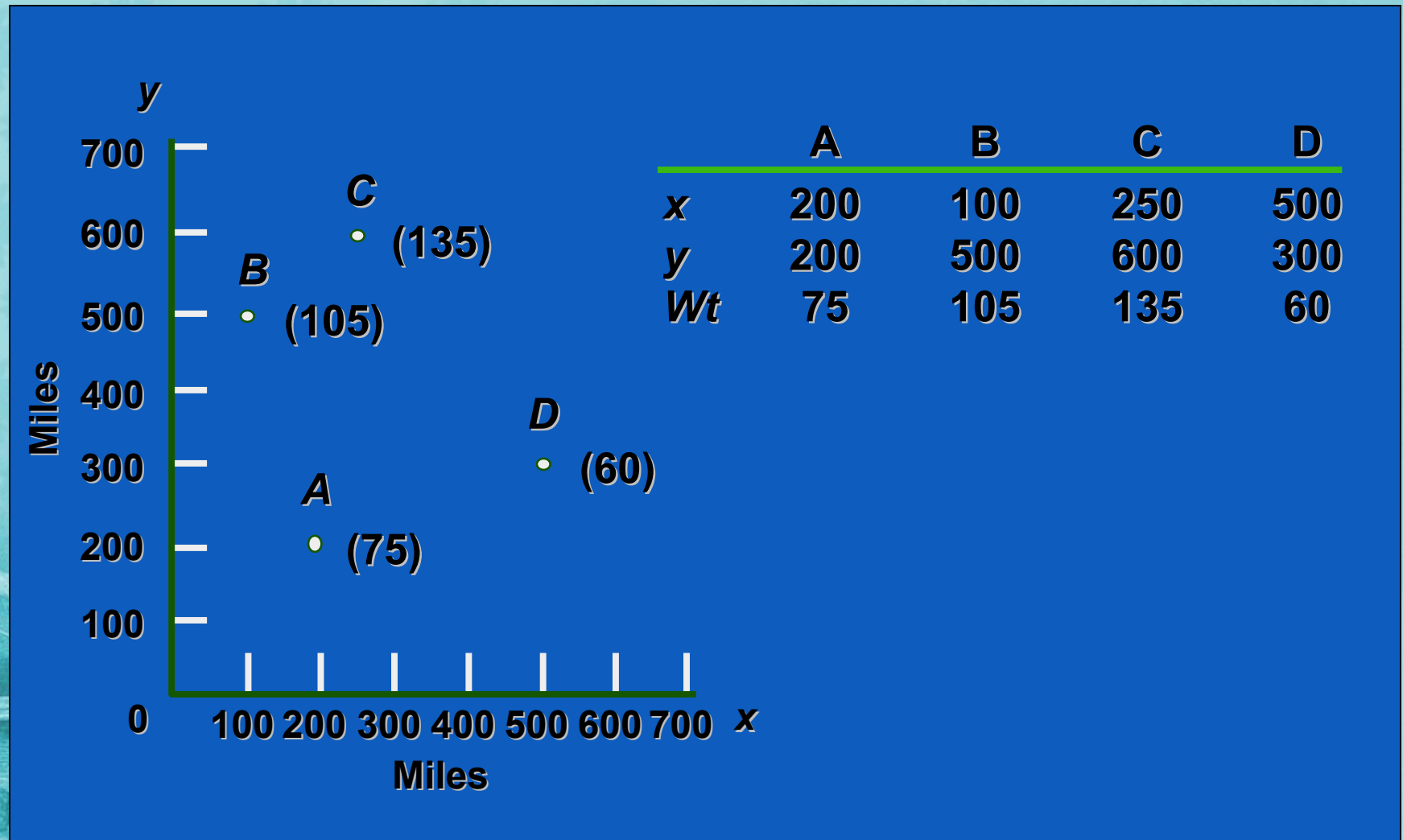
where,

x, y = coordinates of new facility at center of gravity

x_i, y_i = coordinates of existing facility i

W_i = annual weight shipped from facility i

Center-of-Gravity Technique: Example

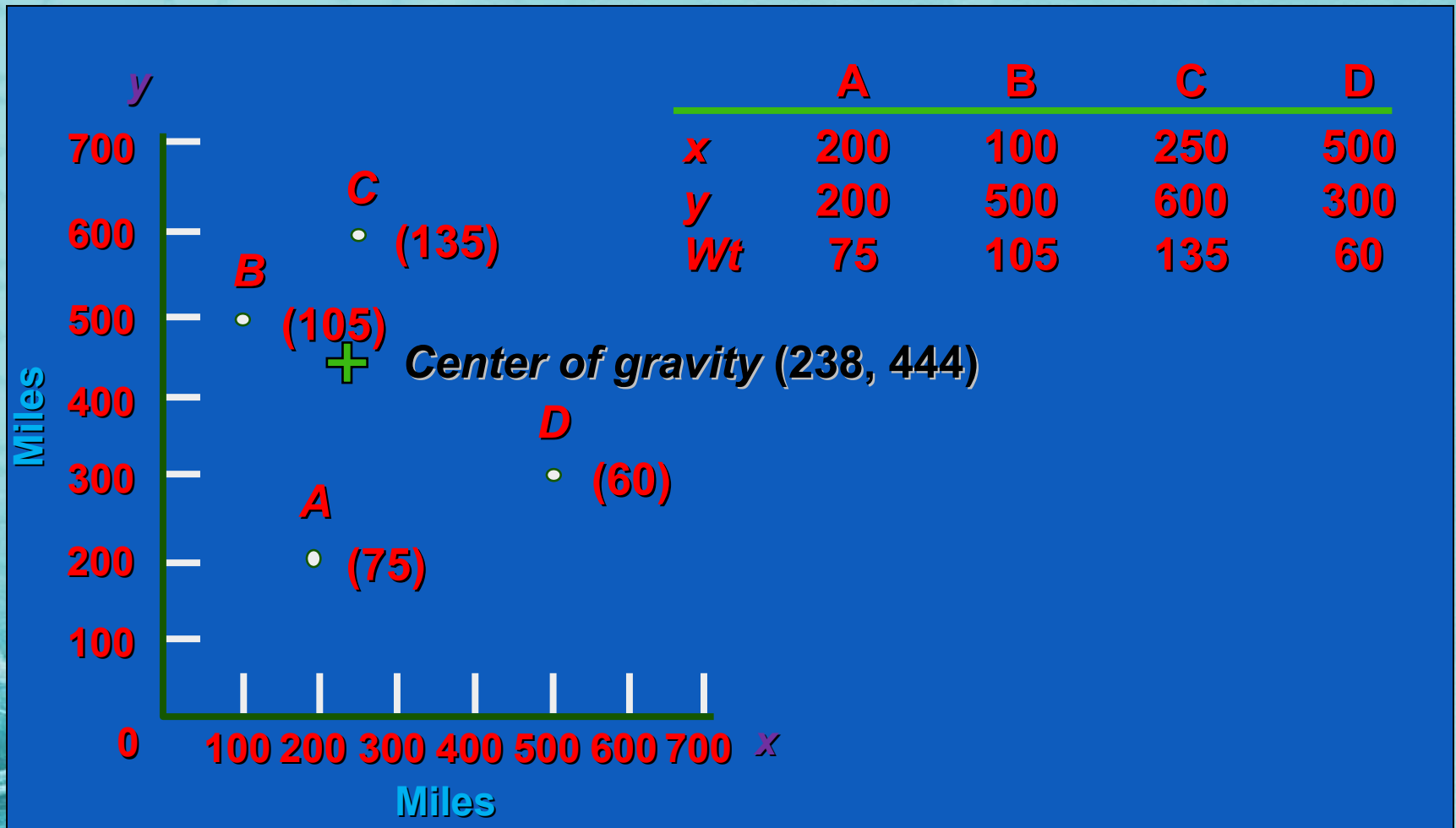


Cont...

$$x = \frac{\sum_{i=1}^n x_i W_i}{\sum_{i=1}^n W_i} = \frac{(200)(75) + (100)(105) + (250)(135) + (500)(60)}{75 + 105 + 135 + 60} = 238$$

$$y = \frac{\sum_{i=1}^n y_i W_i}{\sum_{i=1}^n W_i} = \frac{(200)(75) + (500)(105) + (600)(135) + (300)(60)}{75 + 105 + 135 + 60} = 444$$

Cont...



Load-Distance Technique

- Compute (Load x Distance) for each site
- Choose site with lowest (Load x Distance)
- Distance can be actual or straight-line

Load-Distance Calculations

$$LD = \sum_{i=1}^n l_i d_i$$

where,

LD = load-distance value

l_i = load expressed as a weight, number of trips or units being shipped from proposed site and location i

d_i = distance between proposed site and location i

$$d_i = \sqrt{(x_i - x)^2 + (y_i - y)^2}$$

where,

(x,y) = coordinates of proposed site

(x_i, y_i) = coordinates of existing facility

Load-Distance: Example

Suppliers(existing facilities)

	<u>A</u>	<u>B</u>	<u>C</u>	<u>D</u>
X	200	100	250	500
Y	200	500	600	300
Wt	75	105	135	60

Potential Sites

<u>Site</u>	<u>X</u>	<u>Y</u>
1	360	180
2	420	450
3	250	400

Compute distance from each site to each supplier

$$\text{Site 1 } d_A = \sqrt{(x_A - x_1)^2 + (y_A - y_1)^2} = \sqrt{(200-360)^2 + (200-180)^2} = 161.2$$

$$d_B = \sqrt{(x_B - x_1)^2 + (y_B - y_1)^2} = \sqrt{(100-360)^2 + (500-180)^2} = 412.3$$

$$d_C = 434.2$$

$$d_D = 184.4$$

Cont...

Site 2 $d_A = 333$ $d_B = 323.9$ $d_C = 226.7$ $d_D = 170$

Site 3 $d_A = 206.2$ $d_B = 180.4$ $d_C = 200$ $d_D = 269.3$

Compute load-distance

$$LD = \sum_{i=1}^n I_i d_i$$

Site 1 = $(75)(161.2) + (105)(412.3) + (135)(434.2) + (60)(434.4) = 125,063$

Site 2 = $(75)(333) + (105)(323.9) + (135)(226.7) + (60)(170) = 99,791$

Site 3 = $(75)(206.2) + (105)(180.3) + (135)(200) + (60)(269.3) = 77,555^*$

*** Choose site 3**

Exercise

- 1) An assembly line with 17 tasks is to be balanced. The longest task is 2.4 minutes , and the total time for all tasks is 18 minutes. The line will operate for 450 minutes per day.
 - a. what are the minimum and maximum cycle time?
 - b. What range of output is theoretically possible for the line ?
 - c. What is the minimum number of workstations needed if the maximum output rate is to be sought ?
 - d. What cycle time will provide an output rate of 125 units per day?
 - e. What output potential will result if the cycle time is (1) 9 minutes ? (2) 15 minutes ?

